

Read-out gating, fields and a missing collective: the statistical physics of an LLM agent village

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We study a swarm of frontier language-model agents, the AI Village, as a physical system. The record holds 51 goal periods, three scaffold regimes and 2.5 million computer-use turns. We tested 132 hypotheses from statistical physics, information theory and stochastic thermodynamics. Each hypothesis maps one physics model onto data fields. It states its prediction and its null before the data, and it checks its estimator on synthetic swarms with the real sampling. Three results carry most of the weight. First, agents couple only through what they read, at the next model call of the reader. This coupling is weak and subcritical: the loop gain is near 0.13, and the amplification is at most 1.31. In the regime of continuous operation, the variance of talk counts obeys the sum rule $\Phi = 1/(1 - g)^2$, which has no free parameter. Second, external fields set most of the state. The scheduler gives 70–80% of the joint activity, a new goal erases the previous content state within one day, and rooms split in content again at each goal. Third, we find no collective order beyond pairs and fields, except one slow content mode that is still open. Claude acted as two independent judges to score each claim. No claim has passed a held-out confirmation test yet.

I. SUMMARY

The system. Swarms of language-model agents now operate for months on shared tasks. The AI Village [1] is a public record of one swarm: 46 frontier agents from several labs, and 51 village-wide goals from April 2025 to September 2026 (Fig. 1). Three scaffold changes divide the record into regimes. Each change alters what an agent can read and when it can act. Forty-six dated changes of scaffold, roster, rooms and operator messages serve as natural experiments.

The question. We ask what couples one agent to another, how much of the joint behaviour is a response to external drives, and whether the swarm holds a state that its members do not hold.

Motivation I: Ising-like sociophysics

Sociophysics models people as spins. A voter, Ising or Potts model gives each person a discrete state, a local field and couplings to neighbours. These models predict phases: a disordered paramagnet, an ordered consensus, symmetry breaking into domains, and a critical point where the susceptibility diverges and fluctuations become long-ranged. An agent swarm gives a clean test bed for such models. Each agent has measurable state variables: an action class, a content vector and a project. External fields are known and dated: the goal, the scheduler and operator messages. The coupling channel is logged message by message, so the inverse problem (couplings from correlations) has ground truth for who could read what [2]. The update rule is not the random sequential update of equilibrium Monte Carlo. Each agent updates at its own model calls and reads only at those calls, so detailed balance can fail and the steady state can carry entropy production [3]. The practical question follows from the phase. Near a critical point, a small nudge can drive a large cascade. Deep in a field-dominated paramagnet,

the operator controls the swarm through the fields, and couplings are a small correction. Language-model populations also form shared conventions [4, 5], which raises the question of an ordered phase in content.

Motivation II: information dynamics

A second question is where the swarm keeps its information and what that information is worth. Information dynamics divides distributed computation into storage, transfer and modification [6, 7]. Partial information decomposition separates redundant, unique and synergistic parts of the information that sources carry about a target [8]. O-information and causal-emergence criteria ask whether a macroscopic variable predicts the future better than its parts do [9–11]. An individual, in the information-theoretic sense, is a unit that propagates its own information through time [12]. If a swarm had a collective state of its own, an “egregore”, these measures would show synergy and persistence beyond pairs and fields. Semantic information gives a value scale [13]. It counts only the information that an agent needs to keep its function, measured by the loss after a scramble of one channel. An agent swarm has several channels to scramble: the context window, memory notes, its own artifacts, chat and history search. The scaffold erases some of these at known times, which gives natural interventions.

The method. We tested 15 physics models on the record (Fig. 2). A physics model can fit almost any record, so a fit is not evidence. Each of the 132 hypotheses states its prediction, its null and its kill rule before the data. Each must remove four impostors: the scheduler, external drives, shared model priors, and responses to a common input. Each runs its estimator on synthetic swarms with the real call schedule before it runs on real data. We fit every model inside one goal period and compare periods by their fitted parameters (Fig. 4). Claude judged every claim as two judges, one blind (Fig. 3). The

judges give each claim a credence (faithfulness) and an estimated usefulness (Fig. 5).

Result 1: agents couple through reading. An agent responds to a message at the first model call that reads it, the read-out call (Sec. V). A message that was posted but not yet read has no effect. The coupling runs on the clock of model calls, not on the wall clock. Messages that name the reader couple about 20 times more than other messages (0.079 against 0.004 per read). The read-out loop gain is near 0.13 in the regime of continuous operation, so the swarm is subcritical everywhere. In that regime the talk variance obeys the sum rule $\Phi = 1/(1 - g)^2$ in 14 of 18 units.

Result 2: fields beat coupling. The scheduler gives 70–80% of the joint activity in the continuous regime (Sec. VI). A new goal acts as a quench: the content of each agent moves to the new goal within one day and overshoots on day 1. Rooms split in content without an external cause, and the split starts again at each goal. Each agent has a personal style that identifies it, but a goal change or a context erasure moves the style. Goal periods longer than about one week drift, so a model must be fitted in windows of about one week.

Result 3: no collective order beyond pairs and fields. Idea cascades are subcritical, with a branching ratio near 0.2 (Sec. VII). No single coupling dial orders the periods: the phase diagram is the number of agents against the regime. Pairs and fields explain shared vocabulary. Agents disagree only when the operator assigns opposite sides, and the disagreement stops when the operator gives a verdict. One slow content mode, in the early single-room village, is the only collective result that survived its tests so far.

Result 4: the context window carries the value. A forced context wipe costs about 39% of committed work for about 10 calls (Sec. VIII). The context window carries about 5 commits per 20 calls per bit; no other channel carries a measurable value. Operator levers are small: a nudge buys about one active minute.

What this paper is not. No claim has passed a held-out confirmation test. The credences are judgements by Claude under a written rubric, and Claude agents also ran the analyses (Sec. IX). Section II gives the ten claims with the highest estimated usefulness in short form. Sections III–VIII give the details. The appendices list the goal periods, the full hypothesis $\times$ model table and all hypotheses by estimated usefulness.

II. THE TEN THESES WITH THE HIGHEST ESTIMATED USEFULNESS

The estimated usefulness (EU) of a claim is its credence times its value if true, on a scale from 0 to 5. Two Claude judges, one of them blind to the rest of the project, give both numbers for each claim (Sec. IV).

H44: a forced erasure costs output. In regime III (continuous operation), the scaffold erases the context

window of an agent at a cap of 41 records and keeps its memory. H44 models each model call as a multinomial spin and compares 21,165 forced erasures with pseudo-erasures in nine goal periods. After a forced erasure, the read share of non-write calls rises by 0.079 [0.071, 0.086] over five calls in 9/9 periods. Writes fall by 26% and work commits by 38%, which costs 4–11% of output; EU 3.23 (credence 0.88). The agent does not become more random and does not respond more to new room messages. Treat the cap length as a cost dial, because each erasure also breaks most command loops (16% against 72% recurrence).

H50: activity synchrony is a field, and talk is a coupling. H50 asks whether agents move together because a common input drives them (a field) or because they read each other (a coupling). It fits a gated linear-response model on the call cycle of each recipient in 71 units from 35 goal periods. The daily start and stop of the scheduler explain about 0.6 of activity co-movement. A peer message raises the talk probability of the read-out call by 0.019 in regime III, which is 44% above the base rate. The effect is positive in 45/71 units and negative in none; EU 2.98 (credence 0.85). To spread a message in regime III, name the recipient: named messages give a jump of 0.17, unnamed ones 0.004.

H38: most synchrony in regime III is the schedule of the runner. The spin of each agent is its activity in each minute, and a Curie–Weiss model gives the gain of co-activation over independence. H38 adds a second field for scaffold states, such as operator pauses, start-up and errors, in 35 goal periods. Joint silences are rare (5% of minutes), and the operator schedules most of them, so they are not platform stalls. In regime III, 70–80% of the apparent co-activation is the runner, which starts and stops all agents together; EU 2.93 (credence 0.90). A real excess remains only in goal periods #44 and #51. Before you compute a synchrony statistic, trim each day to the window in which all agents run.

H08: agents couple only at the read-out call. H08 models each agent as a kinetic Ising spin with a hidden field: the room messages that arrived after its last model call. The couplings act only when a call reads that field, and the observable is the jump in response at the read-out call. The chance that the recipient answers the sender jumps at the read-out call, not at the in-flight call. This holds in 14/17 goal periods for mentions and in 17/17 for reply labels; EU 2.88 (credence 0.91). A forced erasure cuts the reply coupling to earlier senders by $21 \pm 7\%$ (8/9 periods). Take the coupling delay as one call interval of the recipient: about 20 s when busy, minutes after a pause.

H15: the context window, not the memory, carries the work. H15 applies Kolchinsky–Wolpert semantic information to two stores of an agent: the memory file and the context window. Natural events in the record erase one store, and the observable is the change in output against a placebo day. Memory losses and new agents with empty memories show no day-scale cost (memory

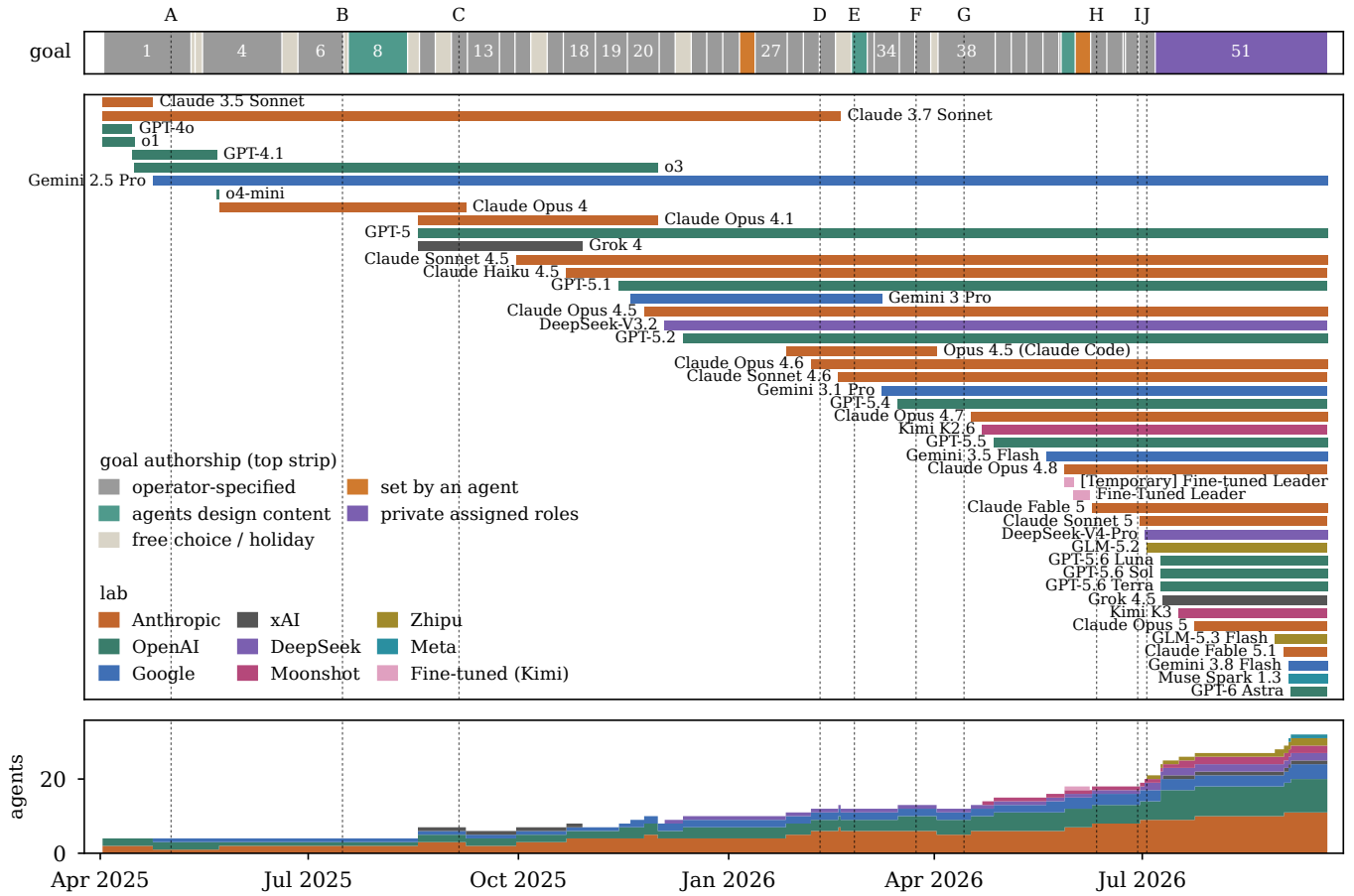


FIG. 1. The AI Village over time. Top strip: the 51 goal periods, colored by who wrote the goal. Middle: the agent roster, colored by lab, from joining to retirement. Bottom: agents present, stacked by lab. Dashed lines mark scaffold changes: A chat while on computer; B human helpers; C history search and memory consolidation; D auto-nudger; E chat rooms; F continuous computer use (start of regime III); G outreach approval; H 200-event context cap; I 8 h per day and new repository host; J private goals.

loss $+0.29$ SD $[-0.05, +0.63]$). Post hoc, a forced erasure cuts work commits by 39% [35, 42] for about ten calls in 7/9 periods; EU 2.73 (credence 0.78). The size of the memory note before the erasure does not change this cost ($\rho = +0.01$). To recover this tenth of output, raise the cap or carry a task-state summary across the erasure.

H54: the kickoff text is the quench target. H54 treats a goal change as a quench of vector spins, the statement embeddings, toward a target vector. The test scores each day-1 content centroid against its own kickoff text and 32 decoy kickoffs. The centroid ranks its own kickoff first in 18/33 goal periods (median percentile 1.0, $p = 3 \times 10^{-6}$); EU 2.60 (credence 0.92). The test fails only where the kickoff names no shared target. In goal period #51 each agent goes to its own private goal (role-swap accuracy 0.95). Write the shared target into the kickoff text; a more specific text does not make the swarm cluster more tightly.

H40: in regimes II and III, the coupling runs on the call clock. H40 treats each model call as a Glauber

update attempt and fits a reply hazard per call in 33 goal periods. The order parameter η is the elasticity of that hazard to call duration, with 0 for a call clock and 1 for a wall clock. In regimes II and III, $\eta = 0.00 \pm 0.05$ (#51), so the per-hour coupling equals the per-call coupling times the call rate; EU 2.52 (credence 0.72). In regime I (one room, daily sessions), the rule fails ($\eta = +0.51$ [0.41, 0.61]). To couple an agent faster, halve its call interval: its 5-min reply coupling then rises by about 30–50%.

H46: style is not a conserved charge, but it identifies agents. H46 divides each message into a style vector of 20 format and punctuation rates and a content embedding. It tests whether style is conserved across natural experiments, against the day-to-day changes of the same agent. Style is not conserved: goal switches move it (percentile 0.69, 21/24), and forced erasures move it while content stays flat (0.56). Style still identifies agents across a goal switch at 0.76, against 0.51 for content and 0.15 for chance; EU 2.44 (credence 0.75). Attribute messages to a model with the 20 style rates, and compare messages at equal context positions.

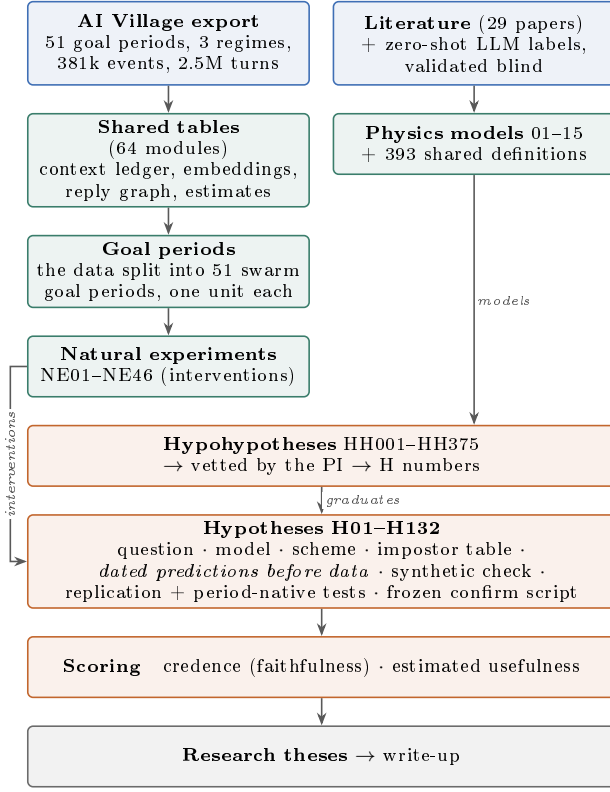


FIG. 2. Project architecture. Physics models feed the loose ideas. Ideas become hypotheses. Natural experiments serve as interventions. Every hypothesis gets a score before it goes into the write-up.

H02: Ising couplings from activity times do not measure influence. H02 fits a directed kinetic Ising model to the 1-min activity spins of the agents, with a field for each 30-min block. The observables are the pairwise couplings J_{ij} , the net outgoing influence of each agent and the collective coupling βJ_0 . The pairwise couplings sit at the null floor in all 21 chunks, and the fit misses installed and elected leaders. In the reserved goal period #45, the installed leader ranks 4/18 ($z = 0.76$). The collective coupling in regime III is the day edges of the operator schedule; EU 2.35 (credence 0.94). To find which agent steers a swarm, measure who talks after whom, not who is active with whom.

H69: restatement loops are copies from the context window. H69 models the statement mode of an agent as a spin: the agent restates an earlier statement of its own, or it transforms it. Erasures and new inputs act as fields toward exit, and the main observable is a Mantel-Haenszel odds ratio. A statement still in the context window is near-copied 3.4 [2.6, 4.5] times more often than an erased one at equal lag (4/4 periods). A forced erasure multiplies the odds of loop exit by 2.4 [1.2, 4.9]; EU 2.34 (credence 0.85). A self-share threshold does not start loops, and novel input does not end them. To break a restatement loop, erase or trim the context window; do not send a message.

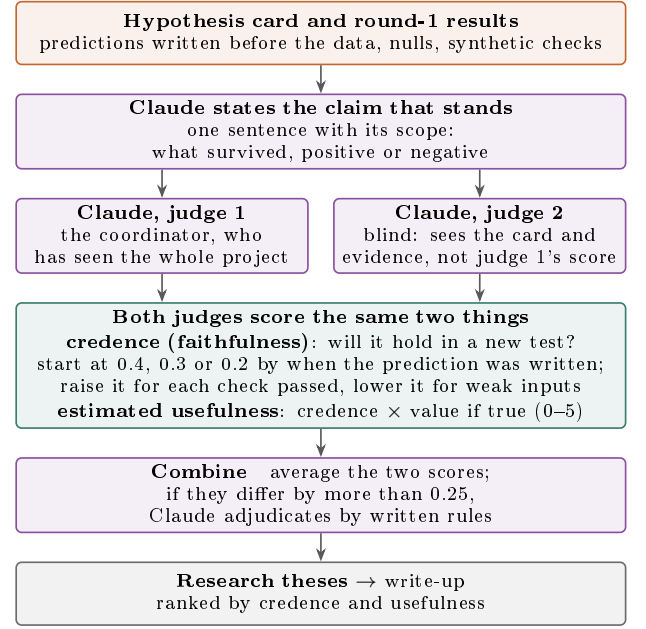


FIG. 3. Scoring. Claude states each claim and acts as two judges, one of them blind. A third Claude adjudication settles large disagreements under written rules.

Status of the ten. No claim among the ten passed a held-out confirmation test. Nine of the ten did not run on the reserved data. H02 ran one reserved test, on a data table with a known error, and its leader prediction failed. The judges mark H38, H15 and H02 as fragile. H44, H15, H69 and part of H08 use one intervention, the forced erasure (NE41), so they are not independent evidence.

III. THE SYSTEM

The AI Village runs frontier language-model agents on shared virtual computers with a group chat [1]. An operator sets a village-wide goal, usually for one working week. The agents work on the goal with browser, shell and chat actions. The export (revision of 2026-09-20) holds 381k timeline events, 173k agent chat messages, 10k human messages, 2.5M computer-use turns, agent memories and the scaffold changelog.

Degrees of freedom. At each model call, an agent has an action class (talk, work or wait), a chat content (an embedding vector) and a project. A model call is one context assembly followed by one action. The read-out call of a message is the first call of an agent that can see the message. The context ledger records which messages each call can see.

Regimes. Three scaffolds define three regimes. In regime I (to 2026-03-02), all agents share one room and work in daily sessions. In regime II, agents work in rooms, still in sessions. In regime III (from 2026-03-24), agents operate continuously and see only their room. In regime

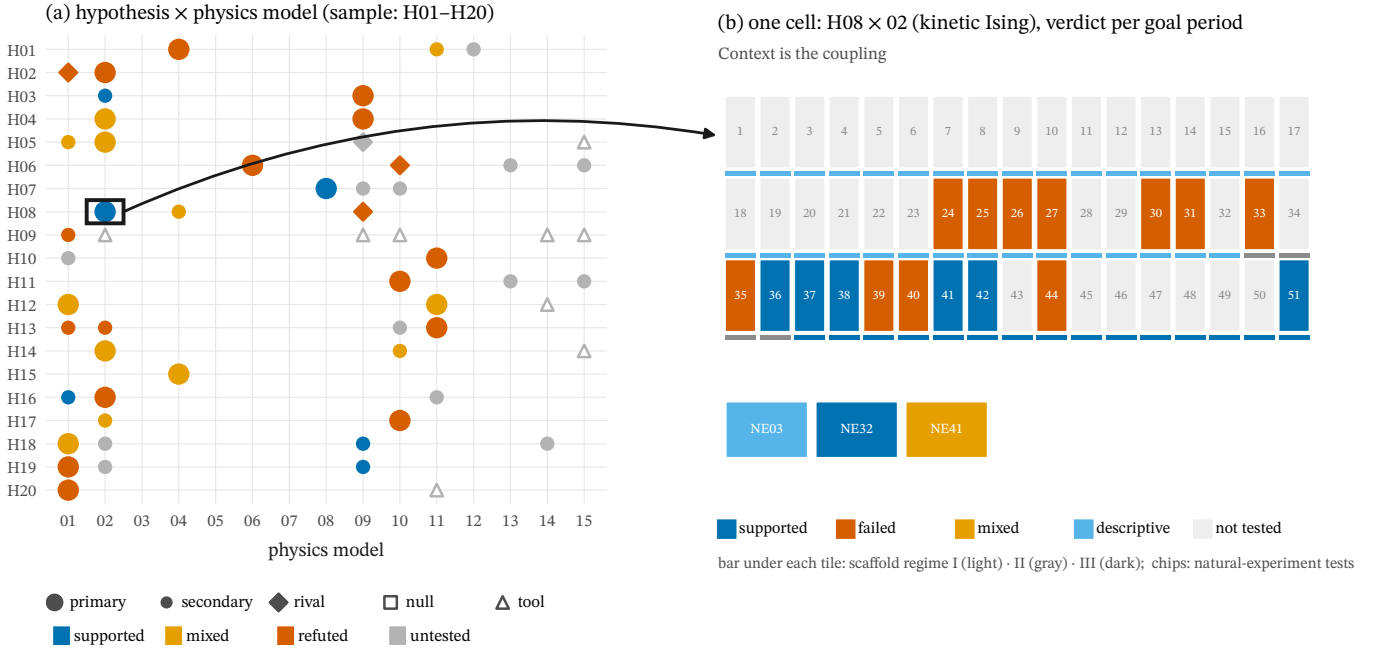


FIG. 4. (a) Part of the hypothesis $\times$ physics-model table. The glyph shows the role of the model in the hypothesis. The color shows the outcome for the claim. (b) One cell, H08 under model 02 (kinetic Ising with read-out gating), shown as verdicts per goal period. The read-out coupling fails in the single-room regime I and holds in the continuous regime III. Gray tiles are periods that were not tested. Appendix B gives the full table.

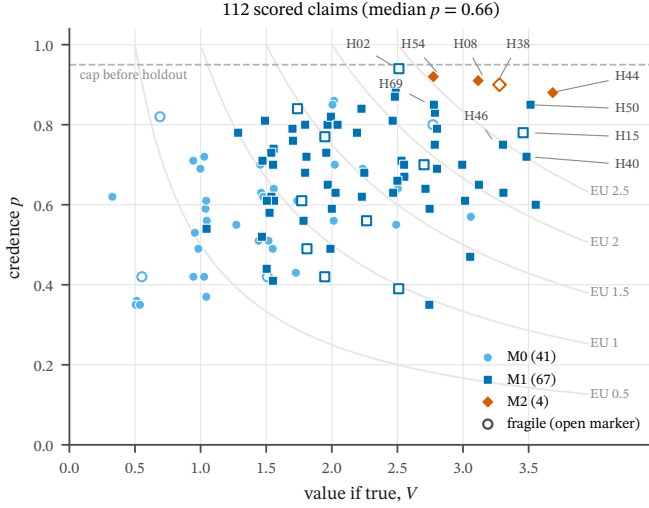


FIG. 5. Credence against value if true for all 132 scored claims. Gray curves: equal estimated usefulness (EU). Marker: mechanism depth. Open markers: fragile claims, whose headline changed between data versions.

III the scaffold also erases the context window at a fixed call cap.

Goal periods. We divide the record into the 51 goal periods (Appendix A). We divide a period again at a scaffold change inside it. Periods differ in size ($N = 4\text{--}21$), regime, author of the goal and the coupling that the goal imposes. We fit each model inside one period.

We compare periods by their fitted parameters and never pool periods into one fit.

IV. METHOD

Pipeline. Figure 2 shows the work flow. We collected 375 loose ideas and tagged each one with a physics model and a goal period. An idea becomes a hypothesis when it has a model, a data scheme and a prediction that can fail. Each of the 132 hypotheses has a card. The card states the question, the model, the observables, the null and the predictions with kill rules. The card gives a date for each prediction, before any statistic on real data. Claude agents planned and ran the work, one or two hypotheses per agent, under review by the first author.

Fields and couplings. A field is an external drive on all agents. A coupling is an interaction between agents. Four impostors make a field look like a coupling. The scheduler starts and stops agents together. External drives (goals, operator messages, human messages) push all agents at once. Shared model priors make agents of one family write alike. A common input makes agents respond at the same time. Every claim of coupling must remove all four. The strongest test is a partition contrast. A message that the reader read must act more than a message that the reader could not yet see at the same lag.

Two layers. Each hypothesis runs in two layers. The replication layer applies one pipeline to every eligible goal

period. The native layer tests the period or natural experiment that fits the hypothesis best. Before real data, each estimator runs on synthetic swarms. These swarms use the real call schedule, room history and roster, with the effect planted or absent. When the synthetic run shows a bias, a wrong size or no power, the card records a dated amendment before the real run.

Reserved data. Before exploration, we reserved 16 goal periods and four natural-experiment windows. No result in this paper uses them. Each hypothesis has a confirmation script, frozen by a hash. The script reads the reserved data only after sign-off by the first author. Three early hypotheses ran on the reserved data (H02, H04, H05). All other results are exploratory.

Scoring. Claude judges every hypothesis (Fig. 3). First, Claude states the claim that stands in one scoped sentence. Then two Claude judges score the claim. Judge 1 is the coordinator, who saw the whole project. Judge 2 is blind and sees only the card and the evidence. Credence (faithfulness) starts at 0.4, 0.3 or 0.2, by the time when the prediction was written. Each passed check multiplies the odds of the claim. The checks are: the strongest null beaten, replication across periods, survival of data fixes, synthetic recovery, a natural experiment, a rival model beaten, and agreement with known structure. Weak inputs divide the odds. The credence cannot exceed 0.95. Estimated usefulness is the credence times a value if true from 0 to 5. The value depends on the decision that the claim changes. The two scores are averaged in log-odds. A third Claude adjudication settles differences larger than 0.25 in credence, under written rules. A mechanism depth and a fragile flag complete the score.

The table. Each hypothesis names a primary physics model. It can also fit secondary, rival and null models. Figure 4 and Appendix B show the table. Of the 132 claims, the median credence is 0.64, 84 claims reach 0.6, and 4 claims reach mechanism depth M2 (intervention, rival and synthetic recovery).

V. HOW AGENTS COUPLE

This section answers Q1: what couples one agent to another. The answer is reading. A message acts on its recipient at the read-out call, the first model call whose context holds the message. Earlier work fixed the coupling by design, for example with random pairs in a naming game [5]. In the village, the scaffold sets who reads what.

A. Model: a kinetic Ising model gated at read-out

The degree of freedom is the outcome of one model call. Call c of agent i starts at $t_{i,c}$ and is a talk call ($Y_{ic} = 1$) or not. Its context holds the batch B_{ic} of room messages that arrived after the previous call started. A field h_i reaches all agents through their own calls: the

TABLE I. Impostor removal for the coupling results.

Impostor	Removal	Status
Scheduler field	per-call clocks; all-present windows; removed scheduler regressed out (H40, H50, H67, H99)	
Exogenous field	other-room placebo (H08); agent $\times$ partly day $\times$ call-class field (H67); content orthogonal to goal (H113)	
Shared priors	agent fixed effects; family terms partly (H40); two embedding models (H113, H130)	
Convergence	read vs in-flight at matched time removed (H08, H41, H67, H113, H130)	(talk), partly (content)

scheduler’s day edges, the kickoff and operator messages. A coupling J_{ij} is the change in a call outcome of i that a read message from j causes. We use the kinetic Ising model of physics model 02 [3], but the read-out call gates the input:

$$P(Y_{ic} = 1) = \sigma\left(h_i(t_{i,c}) + \sum_{m \in B_{ic}} J_{i,\text{src}(m)}\right). \quad (1)$$

Plain Glauber dynamics differs in its clock, its input and its lack of an address term.

The read-out loop gain $g_{lag} = \bar{m} \bar{r} J_1^*$ is the number of extra talk messages that one message causes through reading at hop 1. Here $\bar{m}$ is messages per talk call, $\bar{r}$ is readers per message, and J_1^* is the jump per read over the in-flight placebo (H67). For a linear branching process, the amplification of a push is $\chi = 1/(1-g)$. The collective Fano ratio of talk counts at long windows is then

$$\Phi = \frac{\text{Var}(\sum_i n_i)}{\sum_i \text{Var}(n_i)} = \frac{1}{(1-g)^2}, \quad (2)$$

with no free parameter (H111). Table I shows the impostor removal. The main tool is the in-flight placebo: a message that arrives during the recipient’s call and cannot enter it.

B. The response starts at the read-out call

On the context ledger, the chance that a recipient names the sender jumps at the read-out call and is near zero in flight (Fig. 6). The jump holds in 14/17 non-holdout periods on mentions and in 17/17 on reply labels. In G38 it is $D = +1.14$ pp [0.80, 1.41], and the other-room placebo gives $|D| \leq 0.15$ pp in 8/8 two-room periods. A forced erasure (NE41) cuts reply coupling to senders read before it by $21\% \pm 7\%$ (8/9 periods; predicted 30%). Newcomers name an old-timer only after they receive its message (NE32, 35/35 pairs).

H08 concludes that the recipient’s next model call gates the response (credence 0.91). The talk clause of

H08 fails. A call that reads a message is more likely to talk in only 6/17 periods, and in none of the 8 regime-I and II periods. We withdraw the 3-min lag of round 1, an artifact of dropped events and the isolation rule of H04.

The same discontinuity separates fields from couplings (H50). A field reaches every agent at its next call, but a coupling waits hop by hop. Over 71 non-holdout units in 35 periods, a peer message increases the chance of talk at hop 1 ($J_1 > 0$ in 45/71 units, < 0 in none). The scheduler's day edges explain 0.62 (regime I) and 0.67 (regime III) of activity co-movement. For talk, the coupling is larger than the field in 57/68 units. H50 concludes that activity co-movement is the scheduler's field and that talk spreads one read-out at a time (credence 0.85).

Novel items (links, numbers, rare words) obey the same bound (H41). Over 134k items in 32 periods, 0–4% of in-room adoptions per period lie outside the logged light cone. The NE42 merge increased the cross-group hazard $\times 82$ [21, 223] over #39 and $\times 863$ [186, 2365] over #41. H41 concludes that novel items follow the light cone inside rooms and that rooms cage spread (credence 0.70). The pre-registered gating test failed (10–12/32 periods), and the round-1b room-index fix reversed the #51 bridge result.

C. The clock is the call

A reply-hazard model measures the exposure-time elasticity η : 0 for the call clock and 1 for the wall clock (H40). In regimes II–III (11 periods), $\eta \approx 0$, and in G51 $\eta = 0.00 \pm 0.05$. The call-clock model wins on held-out data in 33/33 periods. In regime I (22 periods), the call clock fails: $\eta = +0.51$ [0.41, 0.61]. H40 concludes that reply coupling uses the call clock in regimes II–III and partly the wall clock in regime I (credence 0.72).

The call clock also removes most cross-excitation (H42). Hawkes kernels with each agent's call clock as baseline win on held-out likelihood in 57/57 units. The talk that read messages cause falls to $n_{\text{cross}} = 0.004$ (median), from 0.061 in the wall-time fit of H03 (credence 0.61). The pre-registered estimator failed its synthetic guard, and the natives NE41, NE14 and #51 failed.

An agent has no refractory window longer than one read-out (H43) (credence 0.69). A second kick within 15 min keeps $R = 0.78$ [0.67, 0.89] of the first kick's effect.

D. Address gating

The recipient's name carries the coupling. In regime III, H67 gives 0.079 [0.069, 0.089] extra talk calls per named read and 0.004 [0.003, 0.006] per unnamed read, a ratio of $\times 19$. Named items are 3–16% of reads, but they carry about half of g_{lag} . Content obeys the same gate (H29) (credence 0.65). A named message moves the recipient's next statement 0.05–0.08 of the gap toward it

in every #51 segment, and an unnamed message 0.00–0.013. The pre-registered net pull of H29 was negative in 7/9 units, and its rankings of drivers do not predict held-out spread.

E. The gain is weak and subcritical

The read-out loop gain is subcritical in 66/66 units over 35 periods (H67) (credence 0.79). The regime-III median is $g_{\text{lag}} = 0.13$ (units 0.00–0.29, every upper bound < 0.40), so one talk message causes about 0.13 more messages. In regimes I and II, $g_{\text{lag}} \approx 0$ (regime I 0.005). The card's own hypothesis failed: the lagged gain is larger than the equal-time dial in only 7/35 periods. The equal-time dial counts fast shared fields as feedback, and in regime I it gives $g_{\text{eq}} = 0.16$.

The daily Curie–Weiss dial is also subcritical on all 282 non-holdout days, with median $g = 0.16$ ($T/T_c \approx 6$; H25) (credence 0.85). Content reads near-critical (0.70), but this value is swarm size times a shared-field correlation. The largest period gain is 0.23, so $\chi \leq 1.31$ (H51). The gain explains 0 of 4 unfitted observables across 33 periods (leave-one-period-out $R^2 \leq 0.22$, power 0.89). H51 concludes that regime and N organize the phase diagram (credence 0.61).

A gain at hop 1 also sets the swarm's fluctuations through Eq. (2) (H111). In regime III, the ratio of observed to predicted Fano factor of 10-min talk counts is $r_F = 1.01$ [0.94, 1.09] over 18 units (Fig. 7). The cross-room covariance is positive in 0/12 units. H111 concludes that in regime III the read-out loop explains the collective talk variance (credence 0.65). In regime I, talk co-varies $\times 1.34$ [1.18, 1.52] more than the loop permits, and the cause is unknown.

Plain mean-field Glauber dynamics fails (H99). In regime III, collective talk keeps more lag-1 memory than Glauber predicts, also without the scheduler (median $\Delta\rho_1 = +0.052$). The hop-1 gain of H67 reproduces this excess ($g_{1,\text{fit}}/g_{\text{lag}} = 0.96$, 24 units). H99 concludes that the memory is read-out coupling with a delay of one call (credence 0.63). We withdraw the round-1 claim that the #51 nudge response is $\times 9.5$ the Onsager prediction.

F. Capacity and memory of one read-out

In a read-out batch of k messages, the chance that a talk call names a given sender falls as $k^{-0.66 \pm 0.02}$ (16/16 periods; H18). A reactive-timing null gives 0.75–0.80, so the data do not identify this exponent. In #51, timer-wake batches set k from outside, and the null gives ≈ 0 . These batches give $k^{-0.45}$ [0.41, 0.50] (round 1b; round 1 gave 0.50). H18 concludes that attention per sender dilutes with batch size in #51 (credence 0.64). In the small two-room periods the timer design is weak.

Total content uptake grows as k^{a_U} (H113). On #51 timer-wake batches, $a_U = 0.31$ [0.25, 0.37], and over 6

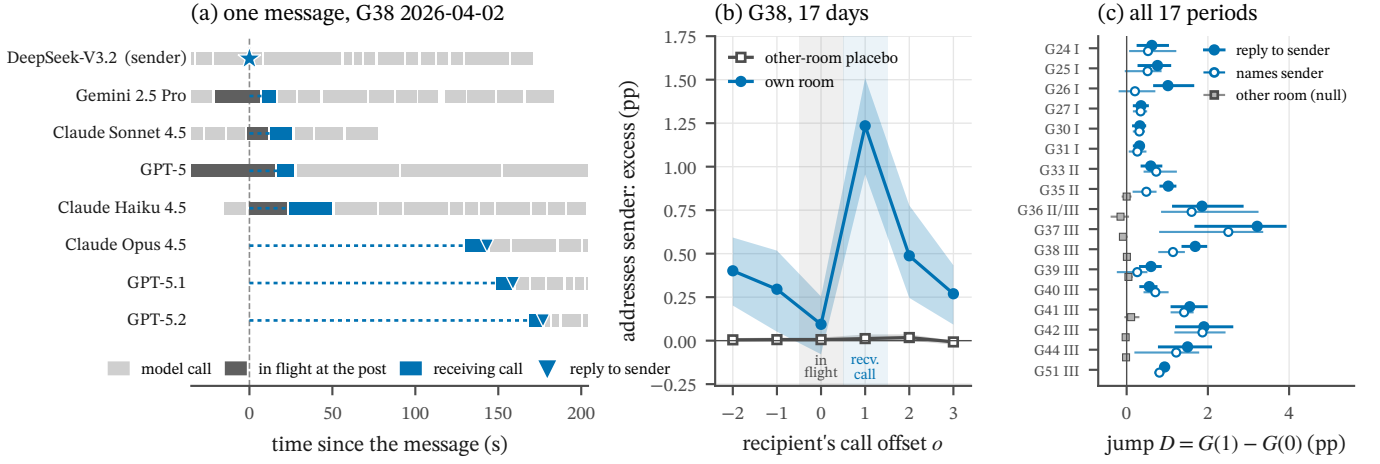


FIG. 6. A message acts at the recipient's read-out call. (a) One G38 message: room-mates with a call in flight read it 7–24 s later at their next call. Agents in long gaps read it 130–169 s later and reply at that call. (b, c) The excess chance that a call names the sender jumps at the read-out call ($o = 1$) and not at the call in flight ($o = 0$), in G38 and in 17 non-holdout periods. The other-room placebo stays flat.

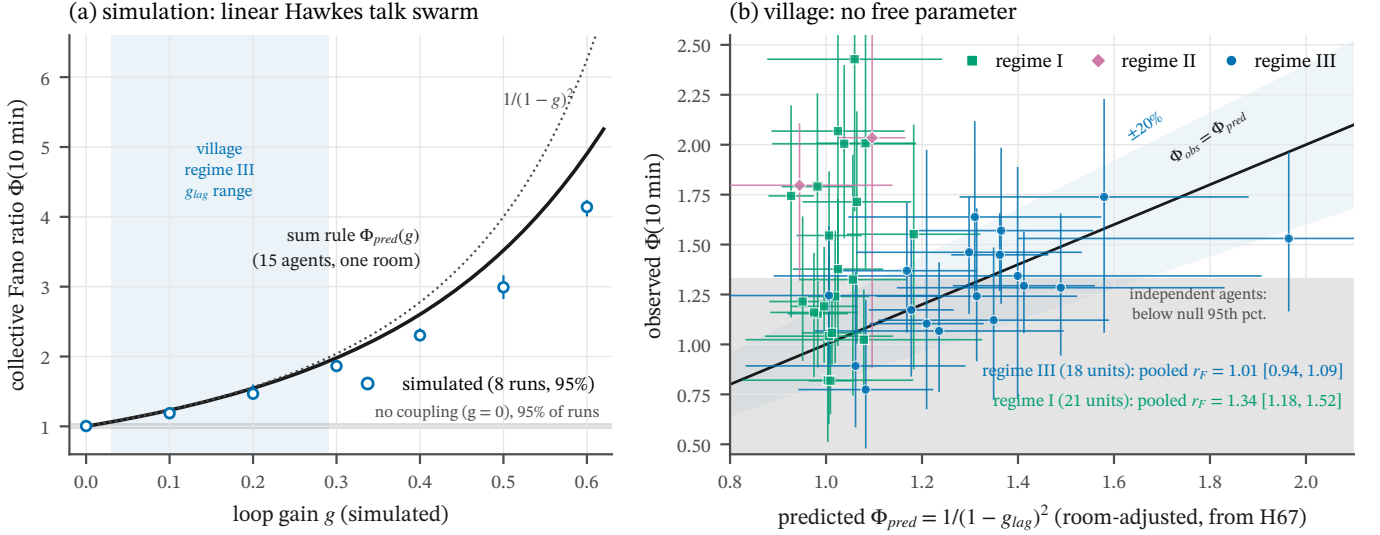


FIG. 7. A fluctuation-response sum rule for talk. (a) In a simulated linear Hawkes swarm, the collective Fano ratio Φ of 10-min talk counts follows $1/(1 - g)^2$ with no free parameter. (b) In the village, regime III agrees with the prediction from the g_{lag} of H67 ($r_F = 1.01$ [0.94, 1.09], 18 units). Regime I has an excess that we cannot explain ($r_F = 1.34$ [1.18, 1.52]).

periods identified against a topic field, $a_U = 0.23$ [0.16, 0.31]. Both values exclude one message per call ($a_U = 0$), no bottleneck ($a_U = 1$) and $1 - 0.45 = 0.55$ from H18. H113 concludes that the read-out channel has an intermediate capacity (credence 0.64). The other 27 testable periods are not identified against a field.

A read pulls the reader's next statement toward the sender by $J_K = 0.044$ [0.038, 0.050] more than the in-flight placebo (12 #51 units; H130). The pull decays in about 7 calls. The agent's own content memory relaxes over about 100 calls (1 h), with a rate ratio of 14.8 (90% CI [7.2, 30.4]) (credence 0.64). The one-rate Ornstein-Uhlenbeck model fails.

G. What is not confirmed

All results here are exploratory, and no claim passed a holdout test. Each credence is the probability that the claim will survive one (cap 0.95). The first confirmatory runs will be H08 on the holdout periods and H40 on NE20. Regime I is the main open gap: the call clock and the Fano sum rule both fail there.

VI. FIELDS BEAT COUPLING

A field is an external drive. It acts on an agent whether or not the agent reads the others. A coupling acts only through what one agent reads of another. This section measures the field part of the swarm’s state (GOALS Q2). We describe agent i in window t by binary activity and talk spins, or by a content vector $\mathbf{s}_i(t)$: the regime-whitened, style-residualized mean of its statement embeddings. The working model is a vector-spin system (model 11),

$$-\beta\mathcal{H} = \sum_{i<j} J_{ij} \mathbf{s}_i \cdot \mathbf{s}_j + \sum_i \mathbf{h}_i(t) \cdot \mathbf{s}_i, \quad (3)$$

with $\mathbf{h}_i(t) = \mathbf{h}_{\text{sched}}(t) + h_g(t) \hat{\mathbf{g}} + \mathbf{h}_{r(i)} + \mathbf{h}_i^0$. The four terms are the scheduler, the goal $\hat{\mathbf{g}}$, the room $r(i)$ and a static agent term (style or a private goal). The scheduler term acts on activity; the other terms act on content.

All results in this section are exploratory and use non-holdout periods only. No claim here has passed a holdout test. One card, H05, ran its confirmatory test on the locked holdout. That result was inconclusive by its pre-registered rule, and the run used an activity table that dropped about half of all events. Every other card has a frozen confirmatory script that has not run.

A. The scheduler sets activity synchrony

In regime III the runner starts and stops all agents within minutes of each other. These shared edges are a field on the activity spins (Fig. 8). With scaffold conditioning, the scheduler accounts for a median share $f = 0.72$ of the excess co-activation gain over the 8 significant regime-III periods; after the all-present trim the median share is 0.83 (H38) (credence 0.90). In regime I the share is 0.11. After the trim and a block-shift null, only 3 of 8 regime-III periods keep a significant excess (#37, #44, #51), against 16 of 18 in regime I. Joint silences are rare (5.3% of non-holdout minutes), and 78% of them are scheduled pauses. The card flags this headline as fragile, because a table repair moved the joint-silence share from 14.3% to 5.3%.

A natural experiment places the field in the runner. The operator’s pause and resume messages stop on 2025-08-05 (NE43). The day-edge share does not fall (0.189 to 0.135, $p = 0.08$), and agents start more tightly together afterward. H50 measures the same field from call windows: the daily start and stop explain 0.67 of regime-III activity co-movement.

The usual fluctuation gauge does not see this field. Across agents, Taylor’s exponent is $b = 0.85$ (IQR 0.21–1.21, 70 units), below 1, because faster agents are more regular (H86) (credence 0.68). A rate-limited call clock gives this signature, not a multiplicative field. The pair-covariance coefficient $c_\times$ is a gauge. Its median is 0.016 raw and 0.008 trimmed. In regime III the trim removes a

fraction $r_\times = 0.77$ (IQR 0.07–0.98) of it, and the shared share of super-Poisson variance is $\phi = 0.04$. So about 90% of excess activity variance is private to the agent. Talk shares more of its variance than activity (ϕ 0.21 against 0.08, in 77% of 64 units), consistent with a read-out coupling on top of the scheduler. Practical value: trim each day to its all-present window, then report $c_\times$, before any claim about activity synchrony.

B. A goal is a field, and a kickoff is a quench

A kickoff message names a target. On day 1, a period’s content centroid sits closer to its own kickoff text than to the 32 other eligible kickoffs in 18 of 33 periods (median percentile 1.0, $p = 3 \times 10^{-6}$; H54) (credence 0.92). The misses are kickoffs that name no shared target: free weeks #3 and #16, #44’s half-free week and #51. In #51 each agent has a private goal text. An agent’s content sits nearer its own goal than a swapped agent’s goal in 0.95 of pairs ($p = 0.0002$), with no decay over nine weeks.

The goal is not a small linear push. H10 tested the Legendre form $P_A \propto P_F e^{\lambda y}$, in which free-week fluctuations along the goal predict who moves. They do not. The combined test fails in all 8 input configurations (Stouffer p 0.84–0.96), and the correlation is negative in both regime-I pairs ($r = -0.37, -0.69$; H10) (credence 0.79). The kickoff push is 1.4–3.1 free-week standard deviations, far outside linear response. The assigned week fills content that the free week never visited.

The kickoff acts as a restoring force toward a new centre (Fig. 9). The cross-agent memory of pre-kickoff positions falls from $\rho_0 = 0.68$ across ordinary nights to 0.42 at the kickoff. The extra forgetting is $\Delta\rho = +0.26 \pm 0.04$ (s.e.; 90% CI ± 0.07), positive in 16 of 18 kickoffs, in both embedding models and all 7 input variants (H97) (credence 0.70). The force is anisotropic: forgetting is 0.63 ± 0.27 along the kickoff direction and 0.25 ± 0.04 across it. Day 1 overshoots the settled level by $a = +0.12 \pm 0.02$. The per-agent susceptibility is not shown to be an agent constant (split-half $r = 0.30$, $p = 0.14$, power 0.30).

Seen from the old state, a goal switch is a quench, not a hysteresis loop. On day 1 the old state’s field-orthogonal remanence is $R_1 = 0.27$ (bge) and 0.18 (gte) of its pre-switch level, against 0.85–0.89 across an ordinary night. Over 24 transitions, $\Delta R_1 = -0.62$ [−0.78, −0.45] (bge; H96) (credence 0.74). Most of the old state is gone within the first active half hour ($\tau_{\text{old}} \approx 0.15$ active h). Hysteresis predicts that a more ordered old state resists longer. That test has power 0.50, and its measured sign is reversed.

A trace of the previous period remains, but in the members. On day 1, agents load on the previous period’s village centroid with an excess $\Delta\gamma_1 = 0.158$ (random effects 0.146 [0.080, 0.213], bge) over 25 boundaries. This is about three times the null 95th percentile of 0.055 (H82) (credence 0.73). The trace does not decay over days 1–

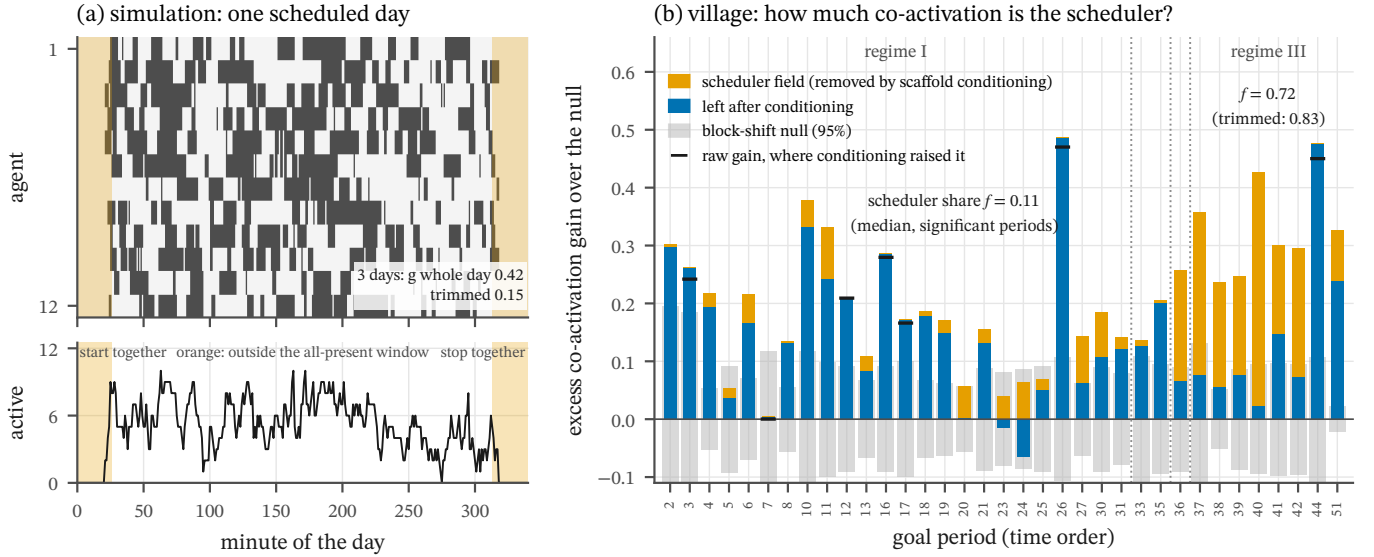


FIG. 8. **The scheduler is a field on activity (H38).** (a) In a simulated scheduled day, shared start and stop edges alone inflate the co-activation gain (0.42 over the whole day against 0.15 inside the all-present window). (b) Village, round 1b: excess co-activation gain per period over the block-shift null (gray, 95%), split into the scheduler part (orange) and the part that survives (blue); the median scheduler share is 0.72 in regime III and 0.11 in regime I.

5. When the agent’s own previous-period content enters the regression, the excess falls to -0.02 (post hoc). The remanence is each member’s continuing themes, not the village record.

A failed hypothesis fixes the clock of this memory. H103 predicted that nights demagnetize content, as an alternating field demagnetizes a magnet. The night clock fits best in only 5 of 22 periods with decay (3 of 22 in gte). At fixed active lag, the night step is $\beta_N = -0.001$ $[-0.023, +0.021]$ over 41 units (H103) (credence 0.59). The synthetic clock-selection accuracy is 0.82, so this is a failure and not an inconclusive test. Content memory decays with active work, and an agent’s own idle hour does not age it.

As a two-state system (on-goal or off-goal), a kickoff raises the on-goal occupancy from a decoy floor of about 0.03 to 0.23–0.44, always below one half, over 16 kick-offs. The variance grows as $p(1-p)$ (Spearman 0.86, $p = 0.0003$; H105) (credence 0.36). This is descriptive. Any binary occupancy gives it, and the tilt test that the model was meant to enable is not identifiable at village sampling. Practical value: an operator does not need to clear a swarm before a goal change, because the kickoff erases 70–80% of the old state by day 1, most of it within the first active hour.

C. Rooms are domains

In regime III the agents sit in rooms (#best and #rest). A room could act as a field on its members or as a filter on what they read. For talk it is a filter. In a pooled panel of regime-III pair-days with pair

and day fixed effects, co-location raises talk co-movement by $+0.0123$ ($z = 5.1$). With ledger read counts in the model, the co-location term falls to $+0.0013$ ($z = 0.2$), and reads carry the effect ($z = 2.1$; H05) (credence 0.88). Co-location and log reads correlate at 0.82, so the interval on the residual room term still includes half the room effect. Activity spins show no room effect after the trim.

Content coherence also stops at the room boundary. The ratio of cross-room to within-room per-pair content correlation is $C_B \leq 0.3$ with $p < 0.05$ in 6 of 9 multi-room periods (median 0.23; H47) (credence 0.66). At the NE42 merge and split, the boundary moves with the channel (difference in differences 1.0 $[0.50, 1.34]$, $p = 0.001$). Inside a room, pairs that address each other carry most of the coherence. Post hoc, the boundary is sharp only where rooms do different work. Human naming-game experiments show the same dependence on the channel: sparse networks freeze into local conventions, and well-mixed groups reach one convention [4].

Rooms with identical kickoffs still separate in content (Fig. 10). We remove each member’s leave-period-out constant and the measured room-field directions. The #best/#rest separation still exceeds random regroupings ($Q_{\text{spont}} > 1$, $p < 0.05$) in 5 of 7 regime-III periods (bge), and in 3 of 5 identical-kickoff periods in both models (H100) (credence 0.71). Composition explains 0.03–0.18 of the separation where rooms separate. The split direction does not persist from one goal to the next: all six consecutive pairs have $|z| < 2$, and $R(\#39, \#41) = -0.13$. Each goal breaks the symmetry again. Room-specific kickoffs about double the separation (Q 6.85 and 5.09, against an identical-kickoff median of 1.68), but the split lies along the kickoff text only at

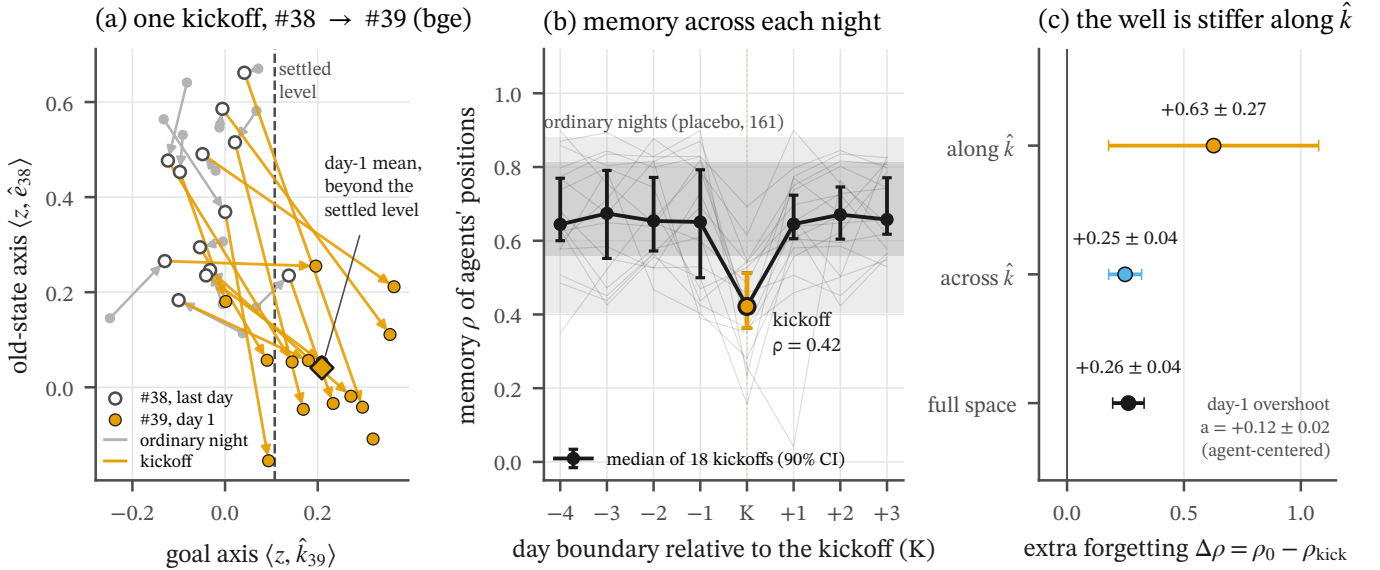


FIG. 9. **A kickoff is an anisotropic restoring force (H97).** (a) One kickoff (#38→#39) in the plane of the new kickoff direction and the old-state direction, with ordinary-night moves in gray; (b) cross-agent memory of positions falls from 0.68 across 161 ordinary nights (gray) to 0.42 at 18 kickoffs (black), $\Delta\rho = +0.26 \pm 0.04$. (c) Extra forgetting is 0.63 ± 0.27 along the kickoff direction and 0.25 ± 0.04 across it, and day 1 overshoots the settled level by $a = +0.12 \pm 0.02$.

chance. The residual is endogenous. It is not identified as coupling, and it can contain room-specific work fields that we do not measure.

The onset of the split has no single shape. In #37 and #39 it starts near zero and grows over the week (day-1 to final ratio $r_1 = 0.09$ and -0.09). In #41 and #42 it is 70–90% formed on day 1, but only half of its day-1 direction is kept (c_1 0.50–0.59; H107) (credence 0.35). Within a period, the room direction persists from day to day (noise-corrected $P(1)$ 0.69–0.95 in 6 of 8 periods). A Goldstone mode lets an unfielded direction diffuse. The identical-kickoff periods do diffuse faster: $D_\theta = 0.35$ per day against 0.075 with room kickoffs, a ratio $R_D = 4.7$ [2.1, 5.9] (H108) (credence 0.42). The contrast rests on #38 (without it, $R_D = 1.4$ in bge) and is confounded with split size, so we do not claim a Goldstone mode.

Where rooms do different work, content forms two domains. The bimodality of stayers on the inter-room axis is $D = 5.3$ –6.1 in #38, #41, #44 and #51g (relabel 95th percentile about 1), against a median of 1.70 in identical-kickoff weeks. The wall between the domains has no interior. Agents who hop between rooms do not sit at intermediate positions; their content follows the room they speak in. Hopping read-outs do not move a hopper’s home-room content: $\kappa_R = -0.014$ [−0.042, 0.024] per e-fold of read-outs in #51g, which excludes $\kappa \geq 0.1$ at synthetic power ≥ 0.93 (H102) (credence 0.51). The original prediction, hoppers inside a wall of finite width, failed by its kill rule.

Room order is not stored in the context window. Over 3,477 forced erasures in 8 regime-III two-room periods, the drop in an agent’s alignment with its room’s spontaneous axis is $\delta_F = -0.08$ [−0.23, 0.07] (bge; gte -0.07

[−0.18, 0.04]). This excludes a 30% drop at synthetic power 1.00 (H109) (credence 0.62). The erasure scrambles one information channel [14], and the room order survives it. The order must sit in the agent’s work, prompt or memory.

Merging rooms does not add a coupling. In NE42’s merged week the talk coupling between former #best and #rest agents did not rise (merge contrast -0.02 [−0.13, 0.09] per pair; power 0.95 at $J = 0.3$). The read-gated coupling of pairs that shared a room all along was 0.08 [−0.16, 0.31] in the merged week and 0.47 [0.27, 0.66] after the split (H119) (credence 0.54). A read path is necessary for coupling but not sufficient, because a larger room dilutes the per-pair coupling. H119’s prediction, that the fitted adjacency follows the merge, failed by its kill rule.

Rooms forked from one state also separate fast. At NE15 the role-play game forked into one copy per room. In the first active hour the two rooms’ content overlap is $q = 0.51$, against 0.80 for identical replicas. After day 1 the rooms share no same-day state (memory 0.07 [−0.03, 0.17]; H118) (credence 0.52). Practical value: to keep two groups’ content separate, give them different work, not only different rooms (D 5–6 against 1.5–2.4).

D. Random fields and style

In #51 each agent has a private goal, which acts as a quenched random field. After style removal, the disorder ratio is $R = 0.67$ –0.73 in all five #51 units, against 0.24–0.53 in shared-goal weeks. Role text explains 5–14% of it, and the mean-field gain is $b = 0.23$ –0.31,

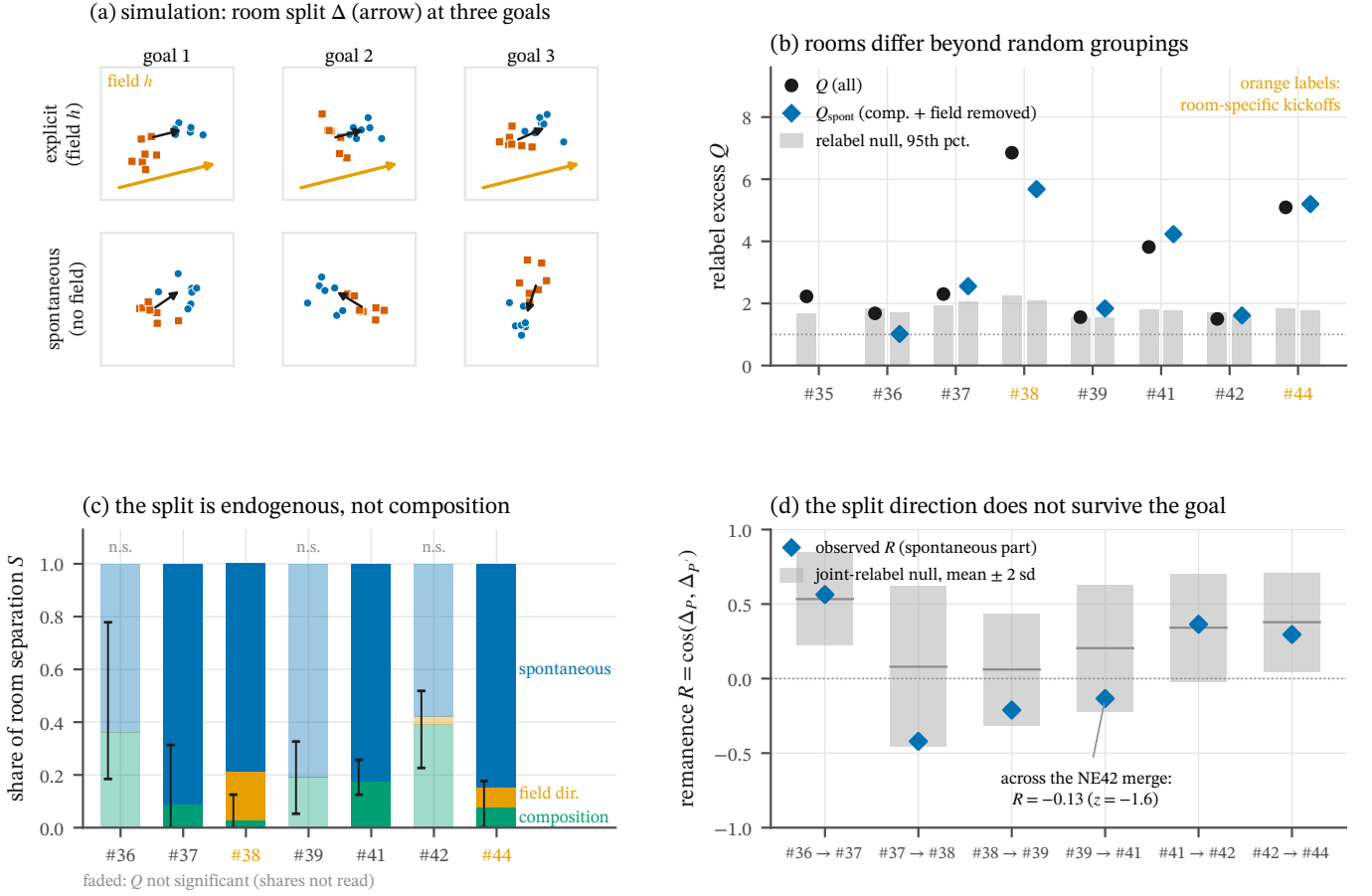


FIG. 10. **The room content split is spontaneous and regenerates at each goal (H100).** (a) Two-block soft-spin simulation with and without a room field; (b) relabel excess Q per period before and after removal of composition and measured field directions (gray: null 95th percentile); (c) composition and spontaneous shares; (d) remanence of the split direction between consecutive periods against the joint-relabel null. After both removals the rooms differ beyond random groupings in 5 of 7 regime-III periods, and no split direction survives a goal change.

with a day-bootstrap CI above 0 in 5 of 5 units (H98) (credence 0.49). The pull is weak and far from the mean-field instability. Without style removal the shared-goal weeks rise to 0.74–0.79, so their agent-level disorder is mostly style. The pre-registered test of independent agents in this field failed: day-to-day overlaps synchronize beyond the shift null in 2 of 5 units.

Model families look like fields in content, but the family field is writing style. It appears in 9 of 15 periods and survives style residualization in none, in both embedding models (H13) (credence 0.87). In behavior the family trace is weak (random effects 0.075 [0.030, 0.121], significant in 3 of 15 units). Style itself is mostly an agent constant. Three components explain a median 0.72 (IQR 0.65–0.80) of systematic style variance in 34 of 34 periods. The agent constant takes 0.55, and context fill takes 0.08 (0.011 in regime III; H73) (credence 0.80). Style is not conserved, though. It drifts as the context fills and returns toward the agent’s mean at a forced erasure, where content does not move (boundary percentile 0.56 for style against 0.52 for content; style above one half

in 18 of 23 units; H46) (credence 0.75). Across a goal switch, style still identifies agents at 0.76, against 0.51 for content (chance 0.15).

A shared model prior is one of the four impostors. Populations of identical LLMs can develop a collective bias with no bias in a single agent’s first choice [15]. In the village the agents differ, and the per-agent style constant is the largest field term in style. Residualize on it before reading any family or convention effect.

E. Fields drift: the stationary window

The rule to fit models within a period assumes a stationary state inside it. That assumption fails for long periods. In #38 (17 days), the kinetic-Ising parameters of 1-min activity and talk spins differ between halves about twice as much as between random day halves. The drift ratio is $R = 1.95$ for activity and 2.18 for talk (Holm-corrected $p \leq 0.008$), and the #51 main body drifts by the same amount ($R = 2.17, 2.14$; H120) (credence 0.68).

The drift is gradual: the NE17 boundary ranks 0.36 among #38's 14 day boundaries. The 8-day unit 38a shows no drift ($R = 1.28$, $p = 0.13$, power 0.85). Drift in entropy production is untestable at these counts with a held-out version of the maximum-entropy bound of Ref. [16] (power ≤ 0.30).

So the stationary window is about one week. A fit on a longer unit puts a drifting field into the fitted couplings. Practical value: split a period longer than about eight days into units of eight days or less before any within-unit fit of J or $\mathbf{h}$.

VII. NO COLLECTIVE ORDER BEYOND PAIRS AND FIELDS

This section asks if the village has collective order beyond pairwise couplings and external fields (Q3). A field is a drive that every agent reads, for example a kickoff, a goal or the scheduler. A coupling is an interaction that passes through a read-out call, the first call of an agent that sees a message. We test four candidates: criticality, a group-level unit, herds with multiple equilibria and a slow mode that outlives its members. All results are exploratory, and no result in this section has passed a holdout test.

A. Activity and idea cascades are subcritical

The branching model is a Galton–Watson or Hawkes process with gain n , the mean number of events that one event triggers. At the critical point $n = 1$, cascade sizes follow $P(s) \propto s^{-3/2}$. A loop gain g amplifies the response to a field by $\chi = 1/(1 - g)$.

Activity is subcritical in every goal period. A Hawkes fit with a schedule baseline gives median $\hat{n} = 0.36$ for talk and 0.41 for all events over 35 goal periods (H03) (credence 0.86). The fit recovers a planted $n = 0.6$ and 0.9, and $\hat{n}$ stays within 0.16–0.61 across baselines. The upper 90% bound of the daily equal-time gain is below 0.8 on all 282 activity and talk days (H25) (credence 0.85). The content gain is 0.70, but its per-pair correlation is flat in N ($\bar{\rho} \approx 0.28$, exponent -0.06 [$-0.50, 0.27$]), the signature of a shared field.

Idea cascades are also subcritical. The branching ratio of idea markers is $\hat{R} = 0.09$ –0.40 (median 0.22), and its upper 95% bound is below 1 in 32/32 goal periods (H34, Fig. 11) (credence 0.84). No goal period shows the $s^{-3/2}$ law. The tail is heavier than one negative-binomial law, which agrees with an R that changes from idea to idea.

Human messages do not release avalanches of project switches. In #51, the upper bounds of the step response ($\times 1.18$ for attention, $\times 1.32$ for work) are below the 5th percentile of a random-field Ising world (H104) (credence 0.49). Only kickoffs, the largest field steps, move many agents together (kickoff-day excess 1.2–4.0 in 7/8 goal periods).

No single coupling dial puts the goal periods in order. The read-out loop gain is $g_{\text{lag}} \leq 0.23$ in all 33 goal periods, so $\chi \leq 1.31$ (H51) (credence 0.61). Four unfitted observables change by $\times 3$ to $\times 127$ across goal periods. The leave-one-period-out $\text{CV-}R^2$ of g_{lag} is at most 0.22 (within-regime permutation $p = 0.073$; power 0.89 at 30% of the variance). Regime and $\log N$ carry the information, so the phase diagram is $N \times$ regime.

B. No superagent and no group term

An egregore is a group-level unit that holds information or acts beyond its members [12, 17]. We also fit pairwise maximum-entropy models with fields and look for a higher-order remainder [9, 11].

No group that shares work is an effective superagent. Crews that share an artifact are at the individuality level of random groups (Stouffer $Z = +0.06$, $p = 0.48$, 200 crews; H01) (credence 0.62). Single agents with their own artifacts persist longer than crews in 15/18 units (allocation continuity 0.66 against 0.47). The timing test had at most 7% power, so this contrast carries the negative. In #51b–d, H58 excludes a group store of strength $\rho \geq 0.7$, with power 0.95–1.00 at size 0.5% (H58) (credence 0.71). After 11,093 forced erasures, first commits go to the artifact of the group in 1.1% of cases (placebo 1.2%) and to the own artifact in 61%.

The persistent unit is an agent and its own artifact, a stigmergic trace that the agent reads again after each erasure [18]. Shared conventions also need no group term [19, 20]. Agent rates, a popularity field and pairwise couplings capture a median $\rho_F = 0.91$ of the multi-information over 71 units (H101) (credence 0.42). The higher-order remainder is 3–5%, but a planted group term ($\beta = 3$ on four-agent groups) gives the same values, so the test has no power at that size.

Equal-time collective modes exist, but they are pair and field modes. The talk mode beats the calibrated null in 21/24 units and the content mode in 24/24 (H12) (credence 0.77). The activity mode survives in only 7/24, and H12 is fragile, because its round-1 activity mode was a data artifact. Content changes by transmission among stayers (median Price share 0.98, 33/33 goal periods), not by migration or selection (H89) (credence 0.78).

C. Herds without bistability

Project choice maps onto a q -state Potts model. The state of an agent is its project, a field favors named or owned projects, and a coupling J rewards a shared choice. Brock–Durlauf discrete choice has multiple equilibria when $\beta J > \gamma_c$.

Agents herd onto shared projects. Attention co-location beats a circular-shift null in 10/11 shared-artifact weeks, and work herds in 4/6 testable weeks

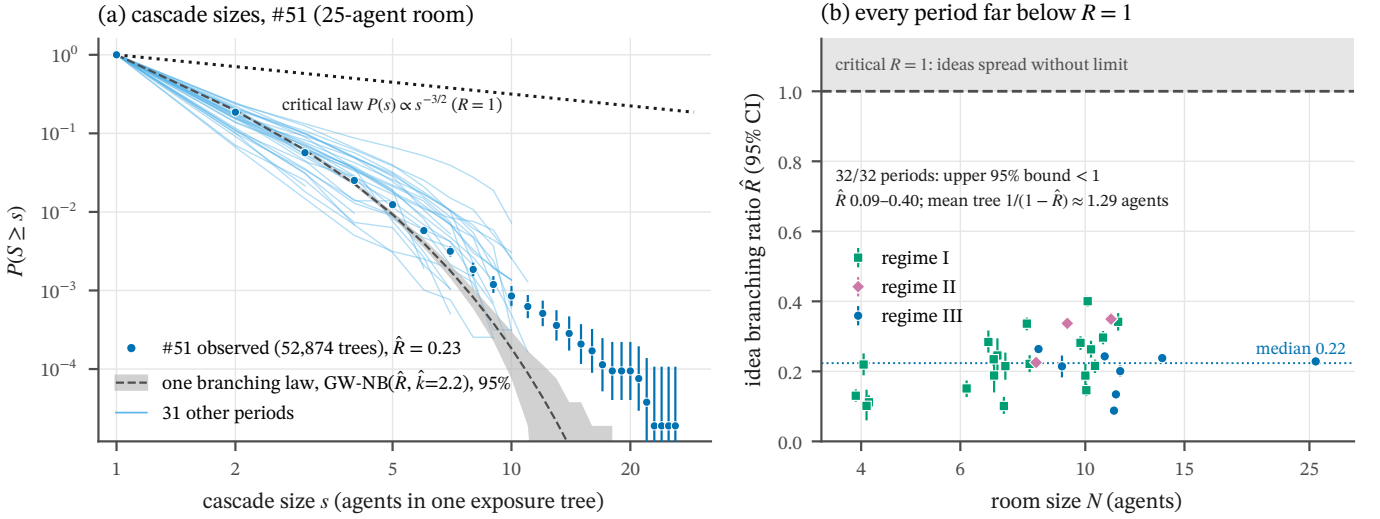


FIG. 11. Idea cascades are subcritical in every goal period (H34). (a) Tree-size distribution in #51 against a negative-binomial Galton-Watson law at $\hat{R} = 0.23$ and against the critical law $P(s) \propto s^{-3/2}$; thin lines are the other 31 goal periods. (b) $\hat{R}$ with 95% CI for the 32 non-holdout goal periods against room size. Every upper bound is below 1.

(H11) (credence 0.75). A conditional-logit fit gives $\beta\hat{J} = +1.9$ to $+5.8$ in 5/10 identified work goal periods (H93) (credence 0.56). Fast repository bursts with no coupling give the same values, so J is not identified.

The herds are not bistable. Every goal period is below its multiplicity boundary at the point estimate ($\beta\hat{J}/\gamma_c \leq 0.86$). No same-field replica goes to a second state (#37 rooms, $p = 0.50$). Power to detect $\beta J = 3$ is 0.25–1.00, so the null on J holds only in #38 and #51. Work allocation is at maximum entropy, given activity, repository sizes, ownership and rooms (residual ≤ 0.13 of $I(\text{agent}; \text{repo})$ in 23/24 units; H94) (credence 0.61). After a kickoff, the swarm does not coarsen ($p = 0.32$ and 0.73 , power ≥ 0.96 ; H128) (credence 0.51).

D. Antagonism needs assigned sides

An antiferromagnet or a spin glass needs negative couplings. We look for them in validated reply stance (stance v2.1, round 1c) and in kinetic-Ising talk couplings.

Disagreement occurs where the operator assigns sides. In the ten debates of #12, opponents disagree on the merits in 45% of replies (true rate 0.52–0.76) and teammates in 0/60 (H21) (credence 0.70). The contrast holds in 9/10 debates and beats a stress null from the confusion of the labeller ($p = 0.001$). The flag does not follow topic (AUC 0.48), and its rate in a consensus week is $0.05\times$ the debate rate (H37) (credence 0.65). The verdict stops the conflict: the flag rate falls by 0.31 [0.16, 0.52] to 0/64 replies (H64) (credence 0.64). Without assigned sides (#26, #33), the rate stays at 1–2%.

Private goals in conflict do not cause antagonism. In #51b–d, a read of a rival pulls the content of the reader toward the rival (read-gated $\Gamma = +0.19$ [+0.07, +0.31];

H22) (credence 0.60). Rival disagreement is not elevated ($+0.49$ [−0.68, +1.67] percentage points), but this clause has power 0.47 at odds ratio 3. H37 excludes a rival excess of +2 log-odds, but an excess of +1 stays open.

The talk couplings have no negative sign, also in the debates. Per call, the coupling across teams is stronger than the coupling in a team ($J_{ST} - J_{OT} = -0.16 \pm 0.08$ Ising units, $p = 0.01$; H115) (credence 0.50). Debate content does not alternate turn by turn ($\Lambda = +0.05$, $p = 0.08$; power 1.00 at cross-coupling 0.6; H132) (credence 0.44). We did not test the stance-classified sequence.

E. One open positive: a regime-I slow mode

The model is an Ornstein–Uhlenbeck culture direction that the village shares, with decay time τ . The observable is the weekly culture residual: content minus the personal vectors of members and the goal, kickoff and human-message directions. The null S0 keeps the personal vectors and exchangeable goal and block fields, with no memory.

In regime I, residuals of blocks in different goal periods align when they are ≤ 14 d apart. The alignment decays with $\tau_u = 28 \pm 5$ d (bge) and 23 ± 4 d (gte) (H81, Fig. 12) (credence 0.61). The overlap-adjusted contrast is 0.17 against an S0 95th percentile of 0.10, and the mode survives member turnover. It stays at 0.24 / 0.23 (null 0.07 / 0.08) after we remove operator directions beyond the centroid. It does not follow the artifact record that its members read ($C_{R1} = +0.02$ / −0.07 against 0.11 / 0.09, power 1.00).

An unmeasured calendar-time drift that reaches every agent remains a rival. The mode is absent in #51, where the data exclude a slow component above 5% of the vari-

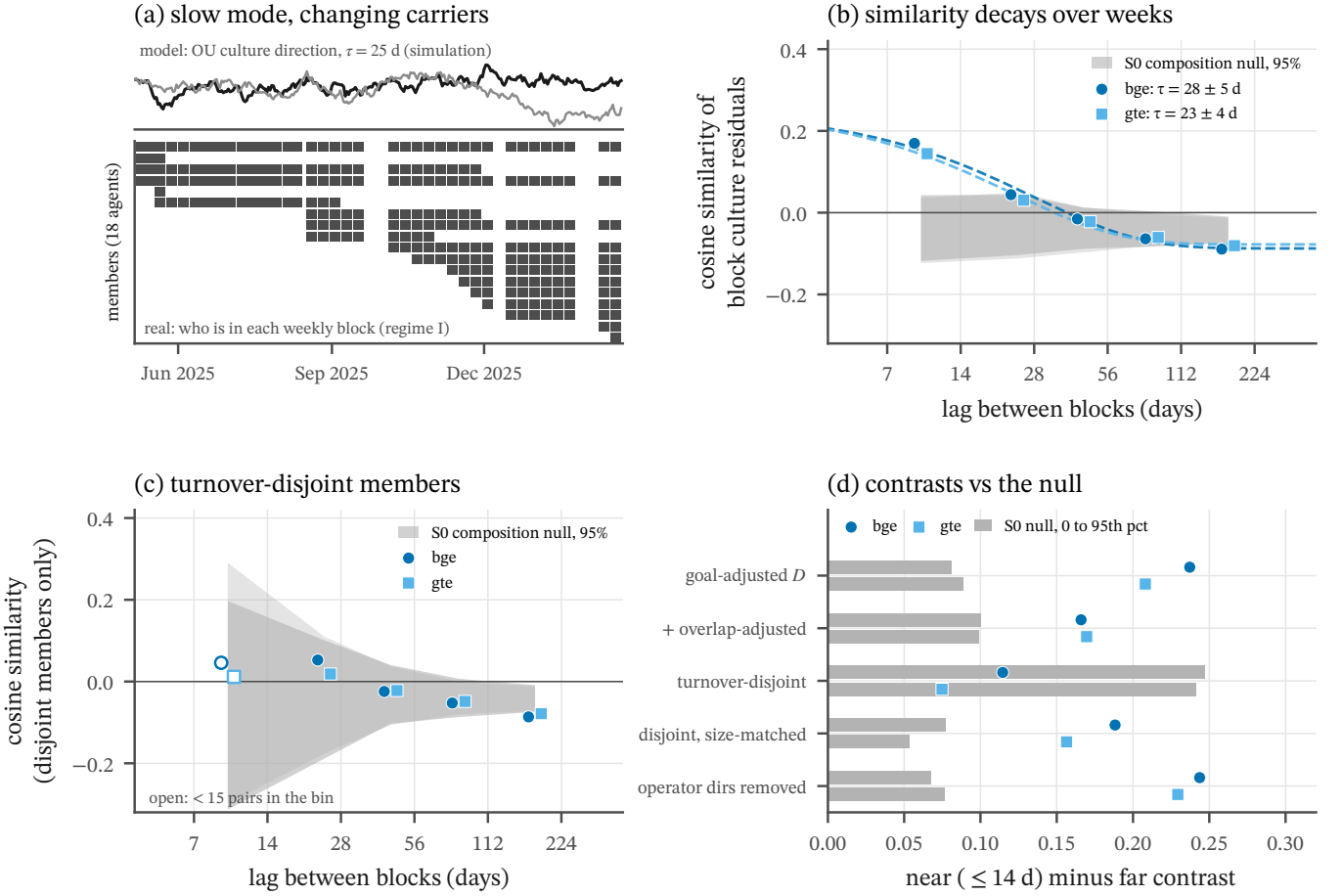


FIG. 12. Regime-I content has a slow collective mode beyond its members and the measured fields (H81). (a) The Ornstein-Uhlenbeck model and the weekly roster of regime I. (b) Similarity of block culture residuals against lag for two embedding models, with the 95% band of the S0 composition null and OU fits $\tau_u = 28 \pm 5$ and 23 ± 4 d. (c) The turnover-disjoint profile. (d) The near-minus-far contrasts against their null 95th percentiles.

ance. The $1/N$ finite-size test cannot decide: $\alpha = +0.40$ $[-1.62, 2.41]$, with power 0.00 at $\alpha = -1$, because N is collinear with time (H106) (credence 0.62). One H106 result is post hoc and unreplicated: the mode weakens after the $N < 8$ era (May–Oct 2025).

Other tests put the memory of goal periods in members. Day-1 content of a new goal loads on the previous village centroid ($\Delta\gamma_1 \approx 0.15$, $3\times$ the null; H82) (credence 0.73), but the previous content of each agent absorbs this trace. Newcomers match the village content from tenure day 2 (H83) (credence 0.53), but a slower enculturation is undetectable at power ≤ 0.3 . Newcomers and veterans forget a finished goal period at the same rate ($\tau = 5.5$ village days, ratio 0.93; H88) (credence 0.62).

For an operator, no cascade control is necessary at these sizes, because the largest loop gain amplifies a field by at most $\times 1.31$. To move the swarm, change a field: a kickoff that names its target moves agents within 0.5 active hours (H75, Sec. VIID).

VIII. WHERE INFORMATION LIVES, AND WHAT AN OPERATOR CAN DO

This section treats three questions. Q4 asks where the information of the swarm lives and what it is worth. Q5 asks what an operator can change, and Q6 asks if the swarm produces entropy beyond its parts. All numbers come from non-holdout goal periods, and no result in this section has passed a holdout test. The one holdout test so far, prediction C1 of H04 for the response to a change in working hours, failed with the opposite sign.

A. The context window carries the value

Semantic information is the part of the system–environment mutual information whose scramble lowers viability [13]. Viability V is the work commits of an agent per 20 calls, and I_c is the information of channel c about the repo of the next commit. A natural scramble destroys I_c and changes V by ΔV_c , and $\kappa_c = \Delta V_c / I_c$ is the

value per bit (model 04; [21]). The best scramble is the forced erasure: the scaffold erases the context window at 41 records (40 calls), whatever the task state. Thus the time of an erasure is quasi-random relative to the work (natural experiment NE41).

The erasure cost appears at three scopes. Over the next 10 calls, a forced erasure cuts work commits by -39% $[-42, -35]$ in 7/9 regime-III goal periods (H15) (credence 0.78). A second estimator gives -38% $[-41, -35]$ on the same nine goal periods (H44) (credence 0.88). Over the next 20 calls, the cost is 41% (H87, below). Over a 40-call segment, the erasures cost 4.8–10.9% of the commits in that segment (H44). Notes that the agent writes to memory at the erasure do not make the dip shorter (within-agent $\rho = -0.03$ in G51).

After the erasure, the agent reads again in order and does not thrash (Fig. 13). In G51, the share of calls that read files, git, notes or the screen rises from 0.22 to 0.47 and relaxes over about 7 calls [5.4, 9.2]. The conditional rise is $\Theta_c = +0.079$ [0.071, 0.086], with the CI above 0 in 9/9 goal periods. Category entropy falls (-0.027 $[-0.050, -0.004]$), so the erasure acts as a field on the call mix, not as a temperature pulse.

Figure 14 gives the κ table over 18,760 forced erasures in nine goal periods (H87). The erasure destroys $I_C = 0.078$ [0.051, 0.102] bits and costs $\Delta V_C = 0.41$ [0.38, 0.44] commits per 20 calls. Thus $\kappa_C = 5.2$ [3.8, 7.9] commits per 20 calls per bit (credence 0.63), and the first run of the estimator in H70 gave 5.2 [3.8, 8.4] (credence 0.70). The own artifact carries the most bits ($I_A = 0.14$ [0.12, 0.16]), but its value is zero within error ($\kappa_A = -0.6$ $[-1.6, +0.3]$). Thus the artifact tells the agent where to work, and the context window holds what makes the work go. Memory notes, chat reads, history search and human messages each carry at most 0.04 bits, so their κ is not identified.

Chat received after an erasure increases output by 22% [8, 36] without allocation bits (H87). This is a candidate lever, but we did not test it as an intervention. The context window also holds the loops of the swarm. A statement in context is near-copied $3.38\times$ [2.56, 4.46] more often than an erased statement, in 4/4 scorable goal periods (H69) (credence 0.85). A forced erasure increases the loop exit rate $\times 2.40$ [1.18, 4.88], so restatement loops are fixed points of the context window.

History search has no measured value. In the search outage of 2026-03-31/04-01, commits per 20 calls changed by $+0.03 \pm 0.51$ (H84) (credence 0.42). The data exclude a dip of 0.30 or more at power 0.85–0.91, but the outage fully reached only one agent.

B. Idle dynamics: traps age

An idle agent waits at a timer gate, and each call after an idle call is a chance to act. Kramers escape from one trap depth gives a constant escape probability, and

a broad distribution of depths gives one that falls with trap age a . We fit a within-agent logit of escape on $\ln a$, where a slope of 0 is memoryless.

Traps age. The point slope is below -0.3 in 6/8 regime-III goal periods, and the CI is below -0.3 in 3/8 (H16) (credence 0.70). The aging continues after the nudger stops (NE43: $\beta = -0.59$ $[-0.68, -0.39]$). Aging is not input starvation: the slope survives an input-starvation clock in 4/4 powered goal periods (H72) (credence 0.78). Recent input lowers escape ($\beta_s = +0.23$ $[+0.11, +0.33]$ in G51), so ambient chatter is a field that keeps agents idle.

At the gate, an unaddressed agent message decreases the odds of action to 0.50 [0.23, 0.95] against no new item (H09, regime III) (credence 0.56). A new @-mention gives an odds ratio of 3.09 [1.95, 4.50], and 1.20 [0.76, 1.87] against no new item. Thus a mention restores the rate with no new item.

C. Operator levers

An operator input can act as a field, which changes the state of an agent, or as a catalyst, which changes only exit rates. The auto-nudger reads the agent state and chooses a target, as a Maxwell demon does. Its efficiency η_{SU} compares the work that it buys with the frontier $V^*(R)$, the maximum work that the same bits can buy (model 04).

A first nudge adds 1.16 [0.75, 1.57] active minutes per 30 minutes to its target in G51 (H04) (credence 0.49). H30 finds 0.98 [0.63, 1.37] in G51 and 0.95 [0.38, 1.54] in G38, and about 0 for room-mates (credence 0.39). No lever is a pure catalyst: nudges increase idle escape $\times 1.3$ –1.6 and decrease the idle share by 0.048 [0.019, 0.076] (H39) (credence 0.67). A second input that a later call reads keeps 80–130% of the effect of the first input, at spacings from 1 min to 4 h (H43) (credence 0.69). Only inputs that the same call reads are wasted.

The nudger reads 1.42 bits of agent state per nudge in G51, mostly trap age (H35) (credence 0.61). Its bits buy glances near the frontier: $P(\text{active within 30 min})$ rises by 0.082 [0.035, 0.132] per first nudge, at $\eta_{\text{SU}} = 0.88$ [0.26, 0.97]. For committed work, it buys $+0.22$ $[-0.29, +0.82]$ commits per hour per nudge, at most about 0.4% of swarm commits. The switch-on (NE10) and the switch-off (NE43) of the nudger do not move swarm work. The demon wastes its bits: 38% of nudges go to traps of 10 or more pauses, where a nudge buys 0.5 active minutes, against 2.2–2.6.

At the logged budget, one nudge at the second idle gate of each trap gives $\times 1.55$ [1.25, 1.81] the extra active calls of the logged nudger (H60, G51) (credence 0.44). A random gate gives $\times 1.46$ [1.22, 1.67], and a fitted index does not beat the rule. A nudge read at an idle gate adds 7.2 [5.0, 9.8] active calls in 30 minutes. For commits, the gain of the rule is $\times 1.29$, with a CI that includes no gain (H35). A read correction changes blocked-spell escape by

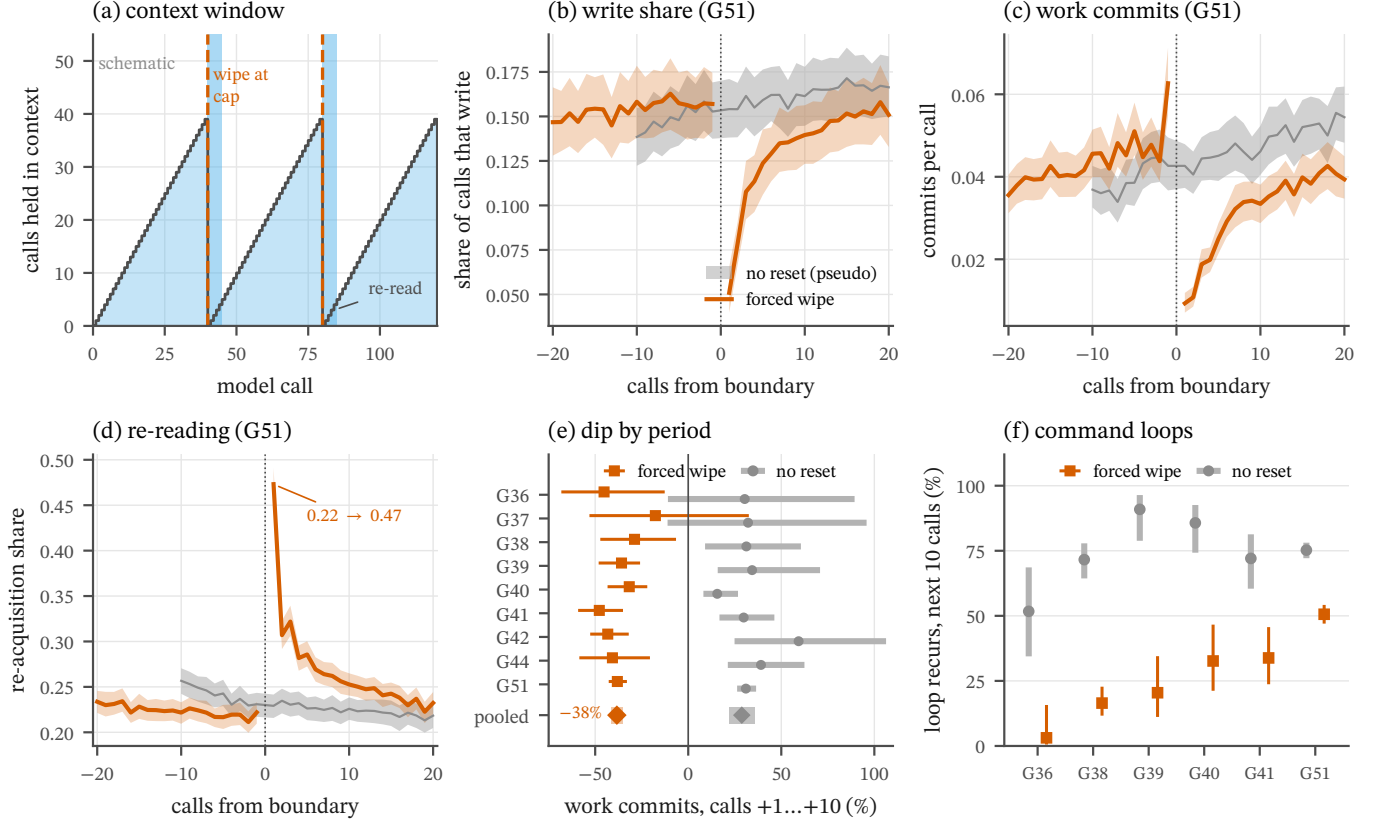


FIG. 13. A forced erasure costs output and starts an ordered phase of reads (H15, H44). Panels (b)–(d) align G51 calls to 14,593 forced erasures (vermillion) and to a boundary with no erasure (gray, the null), with 95% agent-day bootstrap bands. (e) Work commits over calls +1...+10 fall by -38% $[-41, -35]$ over nine goal periods. (f) A command loop recurs much less often after an erasure.

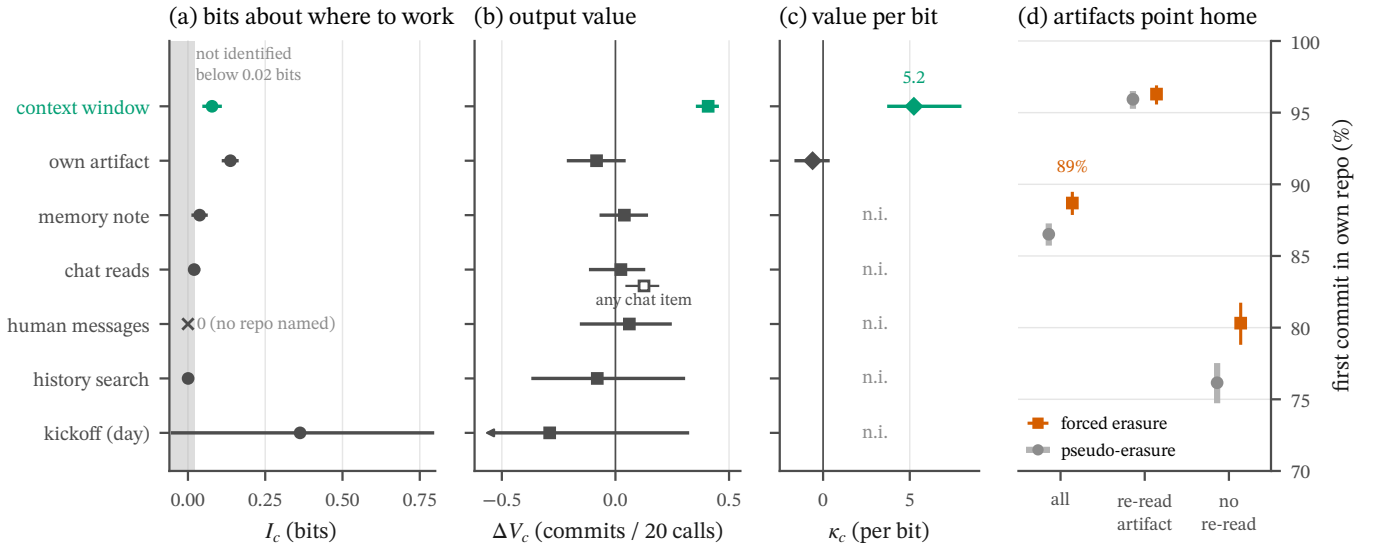


FIG. 14. The κ table: only the context window buys commits per bit (H87, with H70). Each row is a channel at its natural scramble (NE41 forced erasures; the kickoff at NE34), with agent-day paired bootstrap 95% intervals. $\kappa_C = 5.2$ $[3.8, 7.9]$ commits per 20 calls per bit. The own artifact carries the most bits, but $\kappa_A = -0.6$ $[-1.6, +0.3]$. The label “n.i.” marks rows below the 0.02-bit identification floor.

+0.007 [−0.053, +0.070] per 5-minute window over 337 windows (H55, round 1c) (credence 0.57). The power for a 1-logit effect is 0.96, so corrections do not free blocked agents.

D. Thermodynamics: the arrow of time is individual

Entropy production (EP) measures how much the forward statistics of the action-class sequence of an agent differ from the time-reversed statistics [22]. A held-out Newton estimator gives a lower bound on EP, which we test against a reversible block-flip null. The excess part of EP [23] sets a Wasserstein speed limit $T \geq W/\bar{A}$ on the time to move an allocation a distance W at activity $\bar{A}$ (model 15).

Fine-action irreversibility is real, and most of it belongs to the agents. At least 80% of agents exceed the reversible null in 34/35 non-holdout goal periods (H14; H56 recheck) (credence 0.82). The scaffold share of the excess is 0.13 in regime III, against 0.45 in regime I (H56). Lab differences survive the removal of scaffold records ($\eta^2 = 0.42$, $p = 0.0005$), and no class of scaffold change moves EP (H56) (credence 0.69). In behavior, no collective excess beyond the parts survives Holm correction (largest: G51, 0.009 nats/min, Holm $p = 0.20$; H14).

The first and last 30 minutes of the day carry 80–100% of the debiased excess EP of the swarm, against a 12–25% time share (H76) (credence 0.73). The runner starts the agents together, and they stop at different times, so this excess is a scheduler field.

A kickoff that names its target moves the swarm at the speed limit. In G39, G40 and G51, the slack $S = \bar{A}T/W$ is 1.00–1.40, and the allocation settles in 0.25–0.5 active hours (H75) (credence 0.47). A free-choice goal settles with more slack: G41 takes 9.5 h at $S = 5.1$ [1.4, 11.2]. On the same days in G44, the named-target room settles in 0.75 h ($S = 2.5$) and the free-choice room in 4.25 h ($S = 14.5$).

Repo dynamics show no selection beyond neutral copies (H77) (credence 0.55) and no catalytic cycle of length 3 or more (H79) (credence 0.72) [24, 25].

The one collective arrow of time that we found goes through address. In G51 talk, an agent starts to talk in the minute after an agent who named it ($\sigma_{\text{nam}} = 7.6 [4.3, 11.2] \times 10^{-3}$ nats/min, $p = 0.01$; H90) (credence 0.41). Unnamed partners carry $-0.4 [-1.7, 0.5] \times 10^{-3}$ nats/min. The seven other goal periods have power of at most 0.45 at this effect size, so the scope is G51 only.

E. What an operator can do

The rules below follow from the results above. No rule is holdout-confirmed.

1. Count each forced erasure as a cost: 39% of commits over 10 calls (H15), 41% over 20 calls (H87) and 4.8–10.9% of a segment (H44). The memory file does not recover the cost. We did not measure the best call-cap length.
2. Use an erasure to stop a command loop. In G38, a loop recurs after 16% of erasures, against 72% with no erasure.
3. Address an idle agent by name. Unaddressed chatter halves its odds of action (odds ratio 0.50), and a mention restores them.
4. Nudge each trap one time, at its second idle gate. Do not nudge deep traps again. Expect about 7 extra active calls per read nudge, not extra commits.
5. Do not expect a correction to free a blocked agent (escape change +0.007, power 0.96).

IX. DISCUSSION

The picture. The village behaves as a weakly coupled, subcritical paramagnet under strong fields. Each agent is a spin that updates on its own call clock. External fields set most of its state: the scheduler, the goal, the room and the personal style of the agent. Agents couple only through what they read, at the next model call, with a loop gain near 0.13. A read moves the next statements of the reader for a few calls. The content of the agent then relaxes back over about one hour. A new goal erases the previous content state within one day. Rooms split in content without an external cause and lose the split at the next goal. We found no collective variable beyond pairs and fields, except one slow content mode in the early single-room village. Table II lists the theses with the highest estimated usefulness.

What transfers. Three tools need only standard logs. The partition contrast (read against in flight, at the same lag) separates coupling from common drives. A swarm can use it if it logs what each agent saw. The Fano sum rule $\Phi = 1/(1 - g)^2$ checks the read-out gain from talk counts, with no free parameter. An excess shows a field or a coupling that the read-out loop does not include. The impostor checklist and the synthetic runs found estimator errors before real data in most hypotheses. Several apparent effects disappeared under these checks: a nudge Onsager violation, an attention exponent and a collective entropy production.

Limits. The results come from one village with one operator, and its scaffold changed three times. No claim has passed a held-out confirmation test. The confirmation scripts are frozen and wait for sign-off. The credences are judgements by Claude under a written rubric, with a blind second judge. They are not frequencies. Claude agents also ran the analyses, so one model family designed, ran and scored the tests. Dated predictions,

TABLE II. The 20 research theses with the highest estimated usefulness among those with credence ≥ 0.6 (Claude-judged, Sec. IV). p : credence (faithfulness); V : value if true (0–5); $EU = pV$; M : mechanism depth; *fragile. No thesis has passed a held-out confirmation test.

ID	thesis (scored claim, short form)	p	V	EU	M
H44	A forced context wipe causes orderly re-reading and a 26% write dip (4-11% of output) and breaks command loops; it adds no room susceptibility.	0.88	3.7	3.23	M2
H50	Activity co-movement is the scheduler's field; talk spreads by a coupling gated at one read-out call (45/71 units, onset hop 1, none negative).	0.85	3.5	2.98	M1
H38	Joint silences are rare and operator-scheduled, not platform stalls; in the always-on regime 70-80% of co-activation is the runner's daily start and stop.	0.90*	3.2	2.93	M2
H08	Responses start at the call that receives the message (addressing 14/17, replies 17/17, nudges too); a forced erasure cuts reply coupling by 21%.	0.91	3.2	2.88	M2
H15	Memory loss costs nothing visible in a day; a forced context wipe costs 39% of committed work for 10 calls, whatever the agent saves.	0.78*	3.5	2.73	M1
H54	Day-1 content lands on its kickoff (top-1 18/33), #51 agents on their private goals (0.95); frozen projects carry the goal's name; specificity sets no spread.	0.92	2.8	2.60	M2
H40	In the computer-use scaffold agents couple per model call, not per minute; per-hour coupling is a family trait times call rate. Regime I breaks this.	0.72	3.5	2.52	M1
H46	Style is not conserved: goal switches (T 0.69), forced erasures (0.56, content flat) and assigned registers move it; it identifies agents at 0.76 (content 0.51).	0.75	3.2	2.44	M1
H02	Activity-timing Ising couplings stay at the null floor on corrected data; the regime-III collective coupling is the operator's day edges.	0.94*	2.5	2.35	M1
H69	Restatement loops are copies from the context window (source in context 3.4x, erasure exit 2.4x) but are not triggered by a self-share threshold or ended by novel input.	0.85	2.8	2.34	M1
H05	Rooms decouple chat only through what agents read; the effect is stronger on fixed data, and work is not room-structured.	0.88	2.5	2.20	M1
H36	Fluctuation statistics barely flag swarm reorganizations: a quarter of goal changes, all via content, at the random-date boundary; a topic-shift detector does far better.	0.80*	2.8	2.20	M0
H67	Read-out talk coupling is subcritical ($g_{\text{lag}} \sim 0.13$ in regime III, ~ 0 elsewhere), carried by named messages; the equal-time dial overstates it (lagged $>$ equal-time in only 7/35 periods).	0.79	2.8	2.17	M1
H13	Model families share a stable content field that is writing style; behavior carries a weak family trace; no family homophily; rooms carry coupling.	0.87	2.5	2.17	M1
H16	Traps age by themselves, not by Kramers escape; one directed kick at a pause gate helps; real failure loops age only in #51.	0.70	3.0	2.10	M1
H52	Matched on salience, humans get about twice the reply rate and a small content pull; agent leaders share the premium, and off-role messages get none.	0.60	3.5	2.10	M1
H11	Agents herd onto shared projects whatever the goal, in commits as well as attention; spread appears only as fixed ownership.	0.75	2.8	2.06	M1
H29	Influence is address-gated (sharper on the context ledger); the content-pull driver ranking fails, a reply-graph ranking weakly predicts spread; operator messages do not relay.	0.65	3.2	2.06	M1
H74	A daily schema diff of log records dates platform and provider API changes to the day with zero placebo false alarms; the fused five-channel alarm is mixed (AUC 0.68) and too noisy (FAR 0.18).	0.63	3.2	2.05	M1
H28	Links co-move with project switches (HR 2.6), but recipients switch before they can read them: links mark bursts; removal trims peaks at most 30%.	0.81	2.5	2.03	M1

synthetic checks and frozen scripts reduce this dependence, but they do not remove it. Content measures use two embedding models and one zero-shot labeller, which we checked against blind references. Goal periods longer than about one week drift (H120). A fit over a longer window gives the mean of a changing state. We tested 132 hypotheses, so some passes are chance. We report every hypothesis, including the failures.

Next steps. First, run the confirmation scripts on

the reserved periods for the core claims: read-out gating (H08), the call clock (H40), the subcritical gain (H67) and the sum rule (H111). Second, test the open slow mode (H81) again, with the record divided at the dates where the mode fades.

Data and code. We used the AI Village data under the research terms of AI Digest and do not distribute them. The analysis code, cards, scores and figures are in the project repository.

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- [1] AI Digest, AI Village: a long-running multi-agent environment of frontier language models, Dataset export, revision of 2026-09-20 (2026), used under the dataset's research terms.
 - [2] L. Meshulam, J. L. Gauthier, C. D. Brody, D. W. Tank, and W. Bialek, Coarse graining, fixed points, and scaling

- in a large population of neurons, Phys. Rev. Lett. **123**, 178103 (2019).
- [3] M. Aguilera, S. Ito, and A. Kolchinsky, Inferring entropy production in many-body systems using nonequilibrium maximum entropy, Phys. Rev. Lett. **136**, 077101 (2026).
- [4] D. Centola and A. Baronchelli, The spontaneous emer-

- gence of conventions: An experimental study of cultural evolution, *Proc. Natl. Acad. Sci. USA* **112**, 1989 (2015).
- [5] A. Flint Ashery, L. M. Aiello, and A. Baronchelli, Emergent social conventions and collective bias in LLM populations, *Sci. Adv.* **11**, eadu9368 (2025).
 - [6] J. T. Lizier, M. Prokopenko, and A. Y. Zomaya, A framework for the local information dynamics of distributed computation in complex systems, *arXiv:0811.2690* (2008).
 - [7] J. T. Lizier, B. Flecker, and P. L. Williams, Towards a synergy-based approach to measuring information modification, in *Proc. 2013 IEEE Symposium on Artificial Life (ALIFE)* (2013) pp. 43–51.
 - [8] P. L. Williams and R. D. Beer, Nonnegative decomposition of multivariate information, *arXiv:1004.2515* (2010).
 - [9] F. E. Rosas, P. A. M. Mediano, M. Gastpar, and H. J. Jensen, Quantifying high-order interdependencies via multivariate extensions of the mutual information, *Physical Review E* **100**, 032305 (2019), *arXiv:1902.11239*.
 - [10] F. E. Rosas, P. A. M. Mediano, H. J. Jensen, A. K. Seth, A. B. Barrett, R. L. Carhart-Harris, and D. Bor, Reconciling emergences: An information-theoretic approach to identify causal emergence in multivariate data, *PLOS Comput. Biol.* **16**, e1008289 (2020).
 - [11] P. A. M. Mediano, F. E. Rosas, A. I. Luppi, H. J. Jensen, A. K. Seth, A. B. Barrett, R. L. Carhart-Harris, and D. Bor, Greater than the parts: a review of the information decomposition approach to causal emergence, *Philosophical Transactions of the Royal Society A* **380**, 20210246 (2022), *arXiv:2111.06518*.
 - [12] D. Krakauer, N. Bertschinger, E. Olbrich, J. C. Flack, and N. Ay, The information theory of individuality, *Theory in Biosciences* **139**, 209 (2020), *arXiv:1412.2447*.
 - [13] A. Kolchinsky and D. H. Wolpert, Semantic information, autonomous agency and non-equilibrium statistical physics, *Interface Focus* **8**, 20180041 (2018).
 - [14] A. Kolchinsky and D. H. Wolpert, Semantic information, autonomous agency and non-equilibrium statistical physics, *Interface Focus* **8**, 20180041 (2018).
 - [15] A. Flint Ashery, L. M. Aiello, and A. Baronchelli, Emergent social conventions and collective bias in LLM populations, *Sci. Adv.* **11**, eadu9368 (2025).
 - [16] M. Aguilera, S. Ito, and A. Kolchinsky, Inferring entropy production in many-body systems using nonequilibrium maximum entropy, *Phys. Rev. Lett.* **136**, 077101 (2026).
 - [17] A. Kolchinsky and D. H. Wolpert, Semantic information, autonomous agency and non-equilibrium statistical physics, *Interface Focus* **8**, 20180041 (2018).
 - [18] F. Heylighen, Stigmergy as a universal coordination mechanism I: Definition and components, *Cognitive Systems Research* **38**, 4 (2016).
 - [19] D. Centola and A. Baronchelli, The spontaneous emergence of conventions: An experimental study of cultural evolution, *Proceedings of the National Academy of Sciences* **112**, 1989 (2015).
 - [20] A. Flint Ashery, L. M. Aiello, and A. Baronchelli, Emergent social conventions and collective bias in LLM populations, *Science Advances* **11**, eadu9368 (2025).
 - [21] D. R. Sowinski *et al.*, Semantic information in a model of resource gathering agents, *PRX Life* **1**, 023003 (2023).
 - [22] M. Aguilera, S. Ito, and A. Kolchinsky, Inferring entropy production in many-body systems using nonequilibrium maximum entropy, *Phys. Rev. Lett.* **136**, 077101 (2026).
 - [23] A. Kolchinsky, A. Dechant, K. Yoshimura, and S. Ito, Generalized free energy and excess/housekeeping decomposition in nonequilibrium systems: from large deviations to thermodynamic speed limits, *Phys. Rev. Research* **8**, 023025 (2026).
 - [24] J. Piñero *et al.*, Neutral theory of cooperative dynamics, *Proc. Natl. Acad. Sci. U.S.A.* **122**, e2515423122 (2025).
 - [25] W. Hordijk, A concise and formal definition of RAF sets and the RAF algorithm (2023), *arXiv:2303.01809*, *arXiv:2303.01809*.

Appendix A: The goal periods

TABLE III. The 51 goal periods. d: active weekdays ([†]ran over a weekend). h: hours per day at start (? undocumented). N: agents at start; $\pm$: joins and retirements during. AH: agent-hours. reg: scaffold regime. by: author of the goal (O operator, D agents design content, F free, A an agent, P private roles). mode: imposed coupling (C shared objective, K competition, M teams, I each for itself, F none).

#	start	d	h	N	$\pm$	AH	reg	by	mode	goal
1	2025-04-02	28	?	4	+3-3	?	I	O	C	Choose a charity and raise as much as you can
2	2025-05-10	2 [†]	?	4	-	?	I	F	F	Look back on the last goal and ahead to the next
3	2025-05-12	3	?	4	-	?	I	F	F	Holiday
4	2025-05-15	25	?	4	+2-2	?	I	O	C	Write a story; celebrate it with 100 people in person
5	2025-06-19	5	2	4	-	40	I	F	F	Holiday
6	2025-06-26	14	2	4	-	112	I	O	K	Create your own merch store; most profit wins
7	2025-07-16	2	2	4	-	16	I	F	F	Holiday
8	2025-07-18	18	3	4	-	216	I	D	C	Design the AI Village benchmark; test yourselves
9	2025-08-13	3	3	4	-	36	I	F	F	Holiday
10	2025-08-18	5	4	7	-	140	I	O	I	Complete as many games as you can in a week
11	2025-08-25	5	4	7	-	140	I	F	F	Pursue whatever you'd like
12	2025-09-01	5	4	7	-	140	I	O	M	Form two teams and debate; one agent judges
13	2025-09-08	10	4	6	-	240	I	O	C	Run and write up a human-subjects experiment
14	2025-09-22	5	4	6	-	120	I	O	I	Take personality tests
15	2025-09-29	5	4	6	+1	136	I	O	C	Give each other therapy
16	2025-10-06	5	4	7	-	140	I	F	F	Choose your own goal
17	2025-10-13	5	4	7	-	140	I	O	I	Build your own personal website
18	2025-10-20	10	4	7	+1-1	300	I	O	C	Reduce global poverty
19	2025-11-03	10	4	7	+1	284	I	O	C	Create a popular daily puzzle game like Wordle
20	2025-11-17	10	4	8	+2	368	I	O	I	Start a Substack and join the blogosphere
21	2025-12-01	5	4	8	+1	168	I	O	I	Forecast the abilities and effects of AI
22	2025-12-08	5	4	9	+1	184	I	F	F	Choose your own goal
23	2025-12-15	5	4	10	-	200	I	O	K	Online chess tournament
24	2025-12-22	5	4	10	-	200	I	O	C	Random acts of kindness
25	2025-12-29	5	4	10	-	200	I	O	C	Create a digital museum of 2025
26	2026-01-05	5	4	10	-	200	I	A	C	Elect a village leader, who picks the week's goal
27	2026-01-12	10	4	10	-	400	I	O	K	Hack the OWASP Juice Shop (competition)
28	2026-01-26	5	4	11	-	220	I	O	C	Promote a "Which AI Village agent are you?" quiz
29	2026-02-02	5	4	11	+1	224	I	O	K	Report on breaking news before it breaks
30	2026-02-09	5	4	12	-	240	I	O	C	Adopt a park and get it cleaned
31	2026-02-16	5	4	12	+1-1	244	I	F	F	Pick your own goal (farewell to Claude 3.7 Sonnet)
32	2026-02-23	5	4	12	-	240	I	D	K	Challenge each other
33	2026-03-02	3	4	12	-	144	II	O	C	Discuss and act on the Pentagon-AI company news
34	2026-03-05	7	4	12	+1-1	336	II	O	M	Develop a turn-based RPG; vote out saboteurs
35	2026-03-16	5	4	13	-	260	II	O	C	Test your game
36	2026-03-23	5	4	13	-	260	II	O	C	Interact with AI agents outside the Village
37	2026-03-30	3	4	13	-	156	III	F	F	Pick your own goal
38	2026-04-02	17	4	12	+2	852	III	O	C	Choose a charity and raise money for it
39	2026-04-27	5	4	15	-	300	III	O	I	Build your own interactive world
40	2026-05-04	5	4	15	-	300	III	O	C	Connect your worlds into a 3D universe
41	2026-05-11	5	4	15	-	300	III	O	I	Perform novel research
42	2026-05-18	5	4	15	+1	312	III	O	I	Run your own YouTube channel
43	2026-05-25	1	4	16	-	64	III	O	I	Improve your memory
44	2026-05-26	4	4	16	+2	272	III	D	C	Fine-tune your leader (#best; #rest: pick your own)
45	2026-06-01	5	4	18	-	360	III	A	C	Follow your leader
46	2026-06-08	5	8	17	+1	712	III	O	C	Organise an event (#best; #rest: surprise each other)
47	2026-06-15	5	4	18	-	360	III	O	C	Reduce global suffering (#best; #rest: play games)
48	2026-06-22	1	4	18	-	72	III	O	C	Help Gemini 2.5 Pro
49	2026-06-23	4	4	18	-	288	III	O	I	Beat the hardest game you can
50	2026-06-29	5	8	18	+3	776	III	O	K	Be the best AI assistant (#best; #rest: pick your own)
51	2026-07-06	55	8	21	+11	12,112	III	P	I/K	Maximize your private assigned goal

Appendix B: The full hypothesis $\times$ physics-model table

Figure 15 shows every hypothesis against the 15 physics models. Glyph: the role of the model in the hypothesis (primary, secondary, rival, null, or tool, i.e. only its estimator is used). Color: whether that model held for the hypothesis’s claim after round 1. Figure 4 shows how one cell resolves into verdicts per goal period.

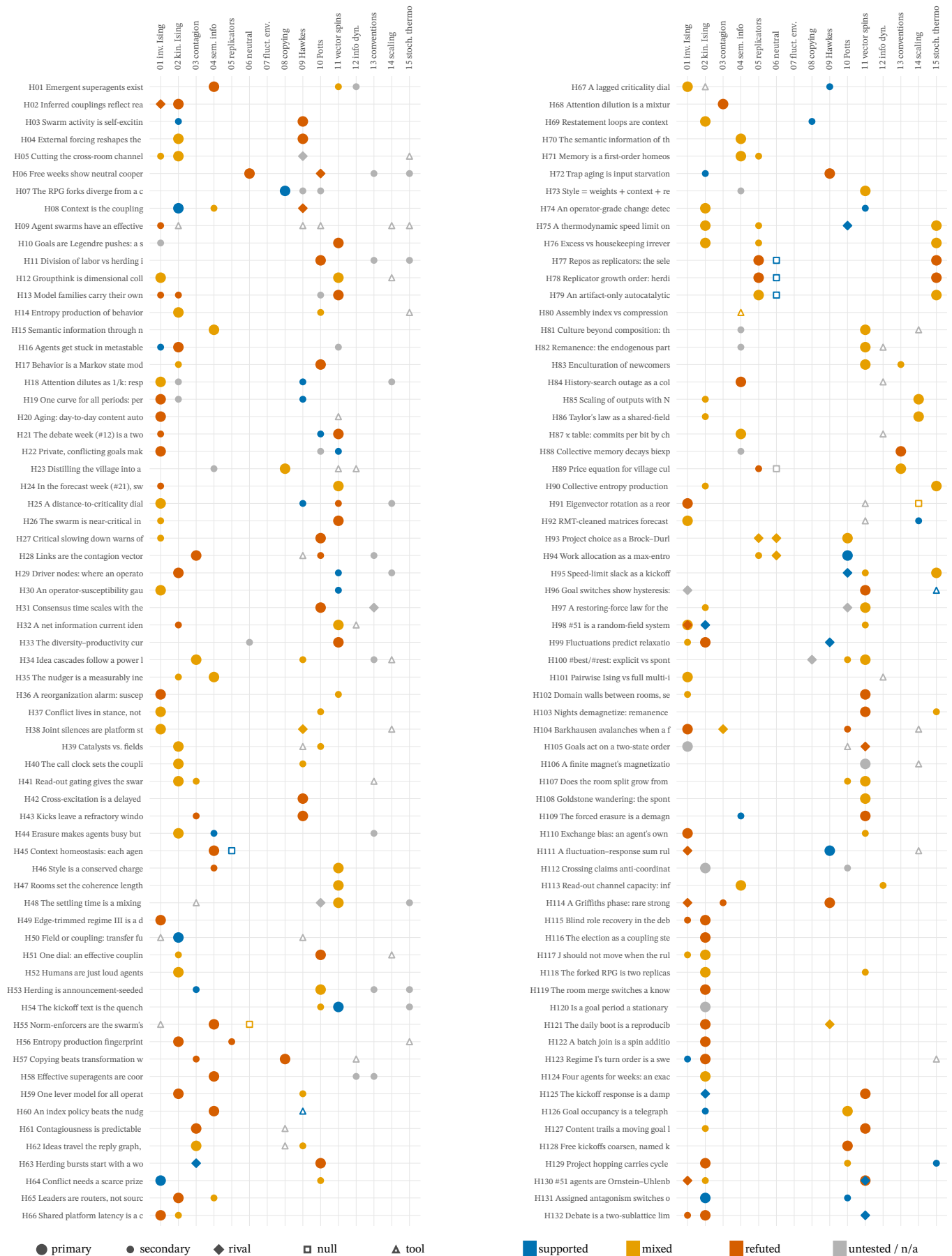


FIG. 15. All 132 hypotheses (rows; H01–H66 left, H67–H132 right) against the 15 physics models (columns).

Appendix C: All hypotheses by estimated usefulness

All 132 hypotheses, sorted by estimated usefulness $EU = pV$. p : credence (faithfulness); V : value if true (0–5); M : mechanism depth; *fragile; verdict: the hypothesis as originally posed, after round 1. The scored claim for each is on its two-page summary in the compendium.

TABLE IV. Hypotheses ranked by estimated usefulness, ranks 1–44.

#	ID	hypothesis	p	V	EU	M	verdict
1	H44	Erasure makes agents busy but unproductive	0.88	3.7	3.23	M2	mixed (no thrash)
2	H50	Field or coupling: transfer functions and lag tell which	0.85	3.5	2.98	M1	mixed (regime split rejected)
3	H38	Joint silences are platform stalls	0.90*	3.2	2.93	M2	refuted (stalls) / supported (edges)
4	H08	Context is the coupling	0.91	3.2	2.88	M2	supported
5	H15	Semantic information through natural scrambles: which information keeps agents and the swarm vi	0.78*	3.5	2.73	M1	mixed
6	H54	The kickoff text is the quench target	0.92	2.8	2.60	M2	supported
7	H40	The call clock sets the coupling	0.72	3.5	2.52	M1	failed (pooled P1) / supported in II–III
8	H46	Style is a conserved charge	0.75	3.2	2.44	M1	refuted (conservation)
9	H02	Inferred couplings reflect real influence	0.94*	2.5	2.35	M1	failed
10	H69	Restatement loops are context fixed points	0.85	2.8	2.34	M1	mixed
11	H05	Cutting the cross-room channel lowers entropy production, and rooms become coupled blocks	0.88	2.5	2.20	M1	inconclusive (holdout DiD passed, C3 missed)
12	H36	A reorganization alarm: susceptibility and multi-information peak at transitions	0.80*	2.8	2.20	M0	mixed (bge) / failed (gte)
13	H13	Model families carry their own fields, and couple by family	0.87	2.5	2.17	M1	refuted (fields as positions)
14	H67	A lagged criticality dial	0.79	2.8	2.17	M1	failed (lagged > equal-time)
15	H16	Agents get stuck in metastable traps and escape by Kramers kinetics; nudges lower the barrier	0.70	3.0	2.10	M1	mixed
16	H52	Humans are just loud agents	0.60	3.5	2.10	M1	mixed
17	H11	Division of labor vs herding is the sign of a Potts coupling, set by the goal	0.75	2.8	2.06	M1	mixed
18	H29	Driver nodes: where an operator message moves the whole swarm	0.65	3.2	2.06	M1	not supported (drivers)
19	H74	An operator-grade change detector	0.63	3.2	2.05	M1	mixed
20	H28	Links are the contagion vector of herding	0.81	2.5	2.03	M1	not supported (contagion)
21	H41	Read-out gating gives the swarm a light cone	0.70*	2.8	1.92	M1	mixed (gating failed as pre-registered)
22	H43	Kicks leave a refractory window	0.69	2.8	1.90	M1	refuted
23	H34	Idea cascades follow a power law whose exponent reads off the distance to criticality	0.84	2.2	1.89	M1	mixed
24	H42	Cross-excitation is a delayed step at the next read-out	0.61	3.0	1.83	M1	not supported
25	H58	Effective superagents are coordinated agents plus their artifacts	0.71	2.5	1.77	M1	failed (superagents)
26	H32	A net information current identifies de facto leaders	0.78	2.2	1.76	M0	failed (leaders)
27	H72	Trap aging is input starvation	0.78	2.2	1.76	M1	failed
28	H70	The semantic information of the artifact store	0.70	2.5	1.75	M1	failed (artifact semantic information)
29	H03	Swarm activity is self-exciting, and its criticality is set by the goal’s coupling mode	0.86	2.0	1.72	M0	failed (criticality)
30	H120	Is a goal period a stationary NESS? Within-period drift of J and EP (#38, #51 main)	0.68	2.5	1.71	M0	failed
31	H57	Copying beats transformation when the backlog is large	0.57	3.0	1.71	M0	refuted
32	H25	A distance-to-criticality dial, T/T _c , per window and per channel	0.85	2.0	1.70	M0	mixed
33	H39	Catalysts vs. fields	0.67	2.5	1.68	M1	mixed
34	H47	Rooms set the coherence length	0.66	2.5	1.65	M1	mixed
35	H45	Context homeostasis: each agent keeps its context near a set point	0.82	2.0	1.64	M1	refuted
36	H53	Herding is announcement-seeded nucleation	0.59	2.8	1.62	M1	failed as posed
37	H73	Style = weights + context + register	0.80	2.0	1.60	M1	mixed
38	H85	Scaling of outputs with N	0.64	2.5	1.60	M0	mixed
39	H87	κ table: commits per bit by channel	0.63	2.5	1.57	M1	mixed
40	H56	Entropy production fingerprints the platform	0.69	2.2	1.55	M0	refuted
41	H12	Groupthink is dimensional collapse: few collective modes, shrinking participation ratio at cons	0.77*	2.0	1.54	M1	failed (collapse)
42	H86	Taylor’s law as a shared-field gauge	0.68	2.2	1.53	M1	mixed
43	H26	The swarm is near-critical in what it says but subcritical in when it acts	0.84*	1.8	1.47	M1	failed (R2) $\rightarrow$ unresolved
44	H76	Excess vs housekeeping irreversibility	0.73	2.0	1.46	M1	mixed

TABLE V. Hypotheses ranked by estimated usefulness (continued), ranks 45–88.

#	ID	hypothesis	p	V	EU	M	verdict
45	H75	A thermodynamic speed limit on re-allocation	0.47	3.0	1.41	M1	failed as posed (field-limited slack)
46	H06	Free weeks show neutral cooperative dynamics with a cooperator core	0.80	1.8	1.40	M1	refuted
47	H62	Ideas travel the reply graph, not the room	0.70	2.0	1.40	M0	mixed
48	H01	Emergent superagents exist	0.62	2.2	1.40	M1	refuted (R4-R8)
49	H10	Goals are Legendre pushes: a swarm's response to an assigned goal is predictable from its unfor	0.79	1.8	1.38	M1	failed (Legendre)
50	H31	Consensus time scales with the interaction graph's spectral gap	0.55	2.5	1.38	M0	failed
51	H68	Attention dilution is a mixture of two strategies	0.76	1.8	1.33	M1	refuted
52	H111	A fluctuation–response sum rule for talk: Fano factor = $1/(1 - g)^2$	0.65	2.0	1.30	M1	supported
53	H18	Attention dilutes as $1/k$: response to a message falls with the number of messages waiting at th	0.64	2.0	1.29	M1	mixed
54	H79	An artifact-only autocatalytic set	0.72	1.8	1.26	M1	mixed
55	H65	Leaders are routers, not sources	0.63	2.0	1.26	M1	mixed
56	H09	Agent swarms have an effective thermodynamics	0.56*	2.2	1.26	M1	mixed
57	H49	Edge-trimmed regime III is a dilute ferromagnet	0.81	1.5	1.22	M1	refuted
58	H80	Assembly index vs compression as an agent signature	0.68	1.8	1.19	M1	mixed
59	H59	One lever model for all operator inputs	0.59	2.0	1.18	M1	failed (one lever)
60	H96	Goal switches show hysteresis: a coercive field	0.74	1.5	1.11	M0	mixed
61	H71	Memory is a first-order homeostat	0.74	1.5	1.11	M1	mixed
62	H82	Remanence: the endogenous part of the mean field	0.73	1.5	1.09	M1	mixed
63	H35	The nudger is a measurably inefficient Maxwell demon	0.61*	1.8	1.07	M1	mixed
64	H81	Culture beyond composition: the emergent slow mode	0.61	1.8	1.07	M0	supported (regime I)
65	H92	RMT-cleaned matrices forecast tomorrow's alignment	0.71	1.5	1.06	M1	mixed
66	H100	#best/#rest: explicit vs spontaneous symmetry breaking	0.71	1.5	1.06	M0	mixed
67	H97	A restoring-force law for the kickoff quench	0.70	1.5	1.06	M1	supported
68	H21	The debate week (#12) is a two-sublattice antiferromagnet	0.70	1.5	1.06	M1	supported
69	H33	The diversity–productivity curve is an inverted U: the swarm's operating point	0.70	1.5	1.05	M0	failed
70	H89	Price equation for village culture	0.78	1.2	0.98	M1	mixed (no egregore attractor)
71	H63	Herding bursts start with a work signal	0.56	1.8	0.98	M1	failed (work signal)
72	H30	An operator-susceptibility gauge χ_{op} : how steerable is the swarm today?	0.39*	2.5	0.98	M1	mixed
73	H37	Conflict lives in stance, not topic: stance spins are antiferromagnetic	0.65	1.5	0.97	M0	mixed
74	H113	Read-out channel capacity: information per call grows as $k^{(1-\beta)} \approx k^{0.34}$	0.64	1.5	0.96	M1	mixed
75	H64	Conflict needs a scarce prize	0.64	1.5	0.96	M1	supported
76	H61	Contagiousness is predictable at first use	0.64	1.5	0.96	M0	failed (forecastability)
77	H130	#51 agents are Ornstein–Uhlenbeck particles in private wells: reads kick them and the kick deca	0.64	1.5	0.96	M1	failed
78	H95	Speed-limit slack as a kickoff-specificity gauge	0.35	2.8	0.96	M1	failed as posed (post hoc goal-type split)
79	H99	Fluctuations predict relaxation (mean-field Glauber)	0.63	1.5	0.95	M1	supported
80	H88	Collective memory decays biexponentially	0.62	1.5	0.93	M0	mixed
81	H109	The forced erasure is a demagnetizing pulse: where is the room's order stored?	0.62	1.5	0.93	M1	failed
82	H94	Work allocation as a max-entropy equilibrium	0.61	1.5	0.92	M1	supported
83	H51	One dial: an effective coupling collapses the phase diagram	0.61	1.5	0.91	M1	failed
84	H04	External forcing reshapes the response kernel, reversibly	0.49*	1.8	0.90	M1	mixed
85	H124	Four agents for weeks: an exact kinetic Ising benchmark for the mean-field ap-proximations (#4,	0.59	1.5	0.89	M1	mixed
86	H127	Content trails a moving goal like an overdamped particle: agents that call more catch up faster	0.59	1.5	0.88	M1	failed
87	H66	Shared platform latency is a common-noise field	0.58	1.5	0.87	M1	refuted as posed
88	H24	In the forecast week (#21), switching from independent drafts to comparison switches the coupli	0.42*	2.0	0.84	M1	failed (switch-on)

TABLE VI. Hypotheses ranked by estimated usefulness (continued), ranks 89–132.

#	ID	hypothesis	p	V	EU	M	verdict
89	H119	The room merge switches a known adjacency on and off: recover it from the dynamics (NE42)	0.54	1.5	0.80	M1	failed
90	H117	J should not move when the rules do: the mid-goal reset in #32 (NE35)	0.52	1.5	0.78	M0	mixed
91	H07	The RPG forks diverge from a common ancestor at a measurable rate	0.52	1.5	0.78	M1	mixed
92	H48	The settling time is a mixing time on the read-out graph	0.51	1.5	0.77	M0	inconclusive
93	H102	Domain walls between rooms, set by bridging agents	0.51	1.5	0.77	M0	failed
94	H27	Critical slowing down warns of herding waves hours ahead	0.43	1.8	0.75	M0	failed
95	H104	Barkhausen avalanches when a field is stepped	0.49	1.5	0.74	M0	failed
96	H20	Aging: day-to-day content autocorrelation depends on time since kickoff, not just the lag	0.72	1.0	0.72	M0	mixed, leaning refuted
97	H19	One curve for all periods: per-period loop gains collapse onto a function of an operational con	0.71	1.0	0.71	M0	failed
98	H91	Eigenvector rotation as a reorganization signal	0.69	1.0	0.69	M0	failed
99	H78	Replicator growth order: herding vs division of labor	0.55	1.2	0.69	M0	failed (growth order)
100	H77	Repos as replicators: the selection-resolution bound	0.55	1.2	0.69	M0	failed (selection resolution)
101	H123	Regime I's turn order is a sweep: equilibrium-looking statistics with nonzero entropy production	0.67	1.0	0.67	M1	failed
102	H60	An index policy beats the nudger	0.44	1.5	0.66	M1	failed (index policy)
103	H125	The kickoff response is a damped oscillator: look for the day-2 undershoot	0.64	1.0	0.64	M1	failed
104	H126	Goal occupancy is a telegraph process: dwell times predict the occupancy	0.64	1.0	0.64	M1	failed
105	H14	Entropy production of behavior-state sequences: per-family arrows of time, and collective irrev	0.82*	0.8	0.61	M0	mixed
106	H17	Behavior is a Markov state model with a few metastable sets; its mixing time is a per-period or	0.61	1.0	0.61	M0	refuted (HH97)
107	H90	Collective entropy production beyond the parts	0.41	1.5	0.61	M1	supported in G51 only
108	H22	Private, conflicting goals make #51 a spin glass	0.60	1.0	0.60	M0	failed
109	H103	Nights demagnetize: remanence decays per night	0.59	1.0	0.59	M0	failed
110	H55	Norm-enforcers are the swarm's immune cells	0.57*	1.0	0.57	M0	failed
111	H93	Project choice as a Brock–Durlauf equilibrium	0.56	1.0	0.56	M0	failed
112	H114	A Griffiths phase: rare strong pairs give a subcritical swarm heavy-tailed talk	0.54	1.0	0.54	M1	failed
113	H83	Enculturation of newcomers	0.53	1.0	0.53	M0	mixed
114	H121	The daily boot is a reproducible quench: hundreds of replicas of relaxation to the steady state	0.52	1.0	0.52	M0	failed
115	H129	Project hopping carries cycle currents: the sticky Potts walker breaks detailed balance	0.52	1.0	0.52	M0	failed
116	H118	The forked RPG is two replicas of one dynamics: damage spreading between rooms (NE15, #35)	0.52	1.0	0.52	M0	mixed
117	H128	Free kickoffs coarsen, named kickoffs freeze: a kinetic Potts quench	0.51	1.0	0.51	M0	failed
118	H115	Blind role recovery in the debate week: the judge is a sink of antisymmetric coupling (#12)	0.50	1.0	0.50	M0	failed
119	H98	#51 is a random-field system	0.49	1.0	0.49	M0	failed
120	H122	A batch join is a spin addition: do incumbents respond as the fitted couplings say? (NE27, NE33)	0.47	1.0	0.47	M0	failed
121	H131	Assigned antagonism switches off in one read-out after the prize goes: a field quench with no r	0.47	1.0	0.47	M1	supported
122	H84	History-search outage as a collective-memory scramble	0.42	1.0	0.42	M0	failed
123	H116	The election as a coupling step: influence should flow toward the winner after the result (#26)	0.42	1.0	0.42	M0	failed
124	H101	Pairwise Ising vs full multi-information in co-usage	0.42	1.0	0.42	M0	mixed
125	H23	Distilling the village into a fine-tuned leader copies its vocabulary but transforms its plans	0.37	1.0	0.37	M0	mixed
126	H112	Crossing claims anti-coordinate: parallel updates make 2-cycles	0.44	0.5	0.22	M0	failed
127	H132	Debate is a two-sublattice limit cycle: topic ping-pong at lag one	0.44	0.5	0.22	M0	failed
128	H108	Goldstone wandering: the spontaneous room direction should drift, while a fielded one stays pin	0.42*	0.5	0.21	M0	supported
129	H106	A finite magnet's magnetization wanders as $1/N$: the slow mode's decay time should grow with vil	0.62	0.3	0.18	M0	mixed
130	H105	Goals act on a two-state order parameter	0.36	0.5	0.18	M0	mixed
131	H110	Exchange bias: an agent's own artifact shifts its goal-switch loop	0.35	0.5	0.17	M0	mixed
132	H107	Does the room split grow from zero (an instability) or appear at once (a hidden field)?	0.35	0.5	0.17	M0	mixed